

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

AND THE

STRUCTURAL OUTLINE

FOR

B.A./B.Sc. (Honours) Course
Second Year Examination, 2011

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STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL AND HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION FOR FIRST YEAR 2010 EXAMINATION (ALREADY HELD) SECOND YEAR 2011 EXAMINATION

The new B.A./B.Sc. (General & Honours) course (at + 3 stage) in the 10+2+3 system of education has been structured keeping in view certain basic guidelines as under :--

1. (i) The scheme should provide a broad-based, interrelated and comprehensive programme of study, built upon the + 2 stage, which ultimately may act as a sound foundation for postgraduate education, participation in competitive examinations, and joining professional courses like those in Law, Education, etc.

   (ii) It is so structured and organised that it is in conformity with the needs of the country in the present context. Apart from providing good general education, it should have an applied component as part of the course work so that it creates awareness of the practical problems facing the society and generates employability of the youth in diverse spheres of activity.

   (iii) It is so designed and formulated that it provides flexibility and relative freedom of choice of courses to the students according to their special interests and capabilities.

   (iv) It is based on intensive teacher–student interaction.

2. The B.A./B.Sc. courses have been so structured and revised that they are comparable both in content and quality.

3. To cope with the fast increasing knowledge in each discipline, it has been considered desirable to cover each elective subject in two papers.

The two-paper scheme will enable the teachers to teach the subject in detail and examine the students accordingly. Effort has to be made to improve the level and adequacy of content of each course in such a manner that the total scheme provides distinct upgradation of the existing curricula for the first degree. In the absence of such an upgradation, the increase in the total educational period from 14 to 15 years for the B.A./B.Sc. will be unjustifiable.

Keeping in view the above objectives and guidelines, the following structural outline has been introduced for the B.A./B.Sc. (General) degree course :

Admissions to B.A./B.Sc. (General/Honours) course for the second and third year (For the students who have passed First Year under this scheme) for the session 2008-2009 (2009 examination) shall be as hereinafter mentioned.

Note : A person who has passed B.A./B.Sc. 1st year or 2nd year examination from other Universities in India may be allowed to migrate to this University subject to the condition that he shall have to clear the deficient subject but the total number of credits required to be earned shall remain the same.

(ii) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

B.A. (General)

The B.A. (General) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2010 Examination (already held)

COMPULSORY

(a) Punjabi – Two Papers — 1 Credit

OR

*History & Culture of Punjab –One Paper — 1 Credit

These papers constitute one subject in each year.

(b) English — 1 Credit

ELECTIVE

+ Any three elective subjects of two credits each including languages and elective vocational subject (To be studied in selected colleges). — 6 Credits

Total : 8 Credits

Environment Education : 50 marks

2nd Year 2011 Examination

Same as in the 1st year

Note :

1. In subjects having practicals, the theory papers and practicals together will be of 2 credit value. The candidate will have to pass in theory and practical/s separately.

2. Each paper of one credit shall be allocated 3 hours of teaching per week during each academic year. However, for subjects having practicals, three hours of teaching will be allocated to each theory paper and two hours for a practical per week or as per requirement of the subject concerned.

*For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2010 examination.

3. B.A. (General) degree holders shall be eligible for admission to the Master's course in any of the elective subjects studied by them during all the three years of the programme of study, earning six credits in each, provided they fulfil the eligibility conditions.

4. A student would offer any science subject, including Mathematics, only if he has passed that subject in the qualifying examination or qualifies in the subject as a deficient/additional subject from the concerned Board/University/Council in the Supplementary Examination subsequent to the admission.

   Provided further that a student can offer :

   (a) Statistics only if he takes up Mathematics.

   (b) Applied Statistics only if he takes up other subject(s) excluding Mathematics.

B.Sc. (General)

The B.Sc. (General) programme of study shall consist of 20 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2010 Examination (already held)

*Punjabi/History & Culture of Punjab	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational Subject (To be studied in selected colleges)	6 Credits
Total :	**7 Credits**

Environment Education : 50 marks

2nd Year 2011 Examination

(a)	English	1 Credit
(b)	Three elective subjects of 2 credits each including Elective Vocational Subject (To be studied in selected colleges) (the same as in the 1st year)	6 Credits
	Total :	**7 Credits**

*For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2010 examination.

(iv) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

Note :

1. The two credits allocated to an elective subject, to be studied in an academic year as spelled out above, shall be covered in two theory papers and a practical or practicals as per requirements of the subject concerned.

2. In the instructional process, it is desirable to strike a balance in the teaching and learning strategies. A close interaction between the teacher and the student is necessary to achieve the above objectives.

3. Each elective subject of 2 credits for the B.Sc. shall be allocated five hours of teaching and four hours of practical work per week during the course of the academic year or as per requirements of the subject concerned.

4. B.Sc. (General) degree holders shall be eligible for admission to the M.Sc. course in any of the three elective subjects which they have studied during all the three years of the degree course, earning 6 credits in each, provided the candidate fulfils the eligibility conditions.

B.A./B.SC. HONOURS

In addition to the main objectives listed in the preamble to the B.A./B.Sc. (General) degree course, the Honours Course seeks to provide course work of advanced nature in one of the elective subjects already being studied from 1st year onwards by a student. Both in content and quality, it would be of a level as gives adequate specialized knowledge in a subject even to make it a culminating point in the programme of study if a student so chooses. Or else, it should provide strong, additional foundation for pursuing the subject at the M.A./M.Sc. level.

Keeping in view the objectives stated in the preamble to the B.A./B.Sc. (General) degree course and that enunciated above, the structural outline for the Honours degree course shall be as under :

B.A. (Honours)

The B.A. Honours Course shall carry 28 credits (each credit carrying the equivalence of 100 marks). Apart from the structure of the programme of study prescribed for the B.A. (General) degree course, credit value of the subjects/papers and hours of teaching allocated to them as already stated in the scheme, the Honours Course shall include four papers of one credit each of an advanced nature in one of the elective subjects a student shall study in all the three years of the degree course.

A student may offer Honours in Second Year in any one of the elective subjects, to be studied by him in all the three years of the course.

Provided that he has obtained at least 50% marks in the subject of Honours in the First Year of the B.A. (General) Course.

For the B.A. (Honours) Course, the distribution of work during the three years will be as under :-

1st Year : The same as in B.A. (General). (8 credits)

2nd Year : The same as in 2nd Year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honour degree.
(8+2=10 credits)

3rd Year : The same as in 3rd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(8+2=10 credits)

Total Credits : 28

A candidate eligible to appear as a private candidate in B.A. Examination may also offer Honours papers in any of the elective subjects offered for B.A. (General) in all three years provided he fulfils the requirement of the Regulation.

B.Sc. (Honours) :

The B.Sc. (Honours) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

A student may offer Honours in any one of the elective subjects to be studied by him in all the three years of the course provided he has obtained at least 50% marks in the subject concerned in the first year Examination of the B.Sc. (General) course.

Of the 24 credits, each student shall offer courses in each year as under :-

1st Year : The same as for First Year of B.Sc. (General). (7 credits)

2nd Year : The same as for 2nd Year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honours degree.
(7+2=9 credits)

3rd Year : The same as for 3rd year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(6+2=8 credits)

Total credits : 24

Both in B.A. Honours and B.Sc. Honours Courses, each paper of one credit shall be allocated hours of teaching, or as per requirements of the subject concerned.

Note : Instruction through audio and/or video cassettes may form a part of Languages course.

(vi) **GUIDELINES REGARDING CONTINUOUS ASSESSMENT OF REGULAR STUDENTS OF B.A./B.SC./B.COM. /B.C.A. COURSES**

IMPORTANT NOTE

(i) In order to incorporate an element of continuous assessment of students, the Colleges will conduct two mandatory House Tests in theory papers – one in the month of September/October and the other in December/January every year.

(ii) *(a)* For September Test, there will be only one paper of one hour's duration in each subject, and for December Test, there will be paper/s on the pattern of annual examination conducted by the University.

There will be a Special Test for those students who could not fulfil the conditions of eligibility. It will not be held to provide an opportunity to all students to improve their earlier score. Those students who are exempted by the Principal of the College from appearing in the House Test/s in September and/or December/January will also be allowed to appear in the Special Test; this Test will determine their eligibility for admission to the examination as well as their score for Internal Assessment.

(b) With a view to meet the grievance of students, if any, on account of scores obtained by them, the answer-books will be shown to them. Difference of opinion on the issue, if any, will be sorted out with the help of respective Heads of departments as well as the Principal of the College.

(iii) Whereas the September House Test will carry weightage of 40 per cent, the December House Test will have weightage of 60 per cent in each subject/paper. The total weightage for both the Tests taken together shall be10 per cent of the total marks in each theory subject/paper. The weightage of 10 per cent marks shall be added to each paper of B.A./B.Sc./B.Com./B.C.A. I, II and III Year which will, henceforth, carry weightage of maximum marks allotted to each paper. A candidate will have to pass in theory and practical/s separately.

(iv) The record of marks secured by the students in the two House Tests will be sent by the respective Colleges so as to reach the office of Controller of Examinations latest by 15th March, failing which the result of the students shall be shown as 'RLA' and the entire responsibility for this would lie with the Principal/s of the College/s.

(v) The Colleges will continue to forward the internal assessment of the students for Practicals. Projects and similar other activities, wherever applicable, to the Controller of Examinations, as usual, so as to reach his office latest by 15th March.

SPECIAL NOTE :

(i) Each theory question paper will be set out of the marks allotted to each theory paper and 10% marks of the maximum marks of each paper will be internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

(iii) It will not be mandatory for the students to separately pass in the internal assessment.

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING IN THE HONOURS SUBJECTS OF THE B.A./B.Sc. SECOND YEAR (THREE-YEAR DEGREE COURSE) EXAMINATIONS, 2011

ENGLISH

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Paper-I : 16TH AND 17TH CENTURY LITERATURE

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Texts :

1.	Shakespeare	:	*Macbeth.*
2.	Congreve	:	*Way of the World.*
3.	John Donne	:	(a) *A Valediction : Forbidding Mourning.* (b) *The Flea.* (c) *The Relique.* (d) *Death Be Not Proud.* (e) *Hymn to God, The Father.*
4.	Sonnets of Milton	:	The following sonnets : *VII Captain or Colonel or Knight in Arms.* *XVI To The Lord General Cromwell.* *XIX On His Blindness.*

Testing :

1.	One question on age/movements with internal choice.	(15 marks)
2.	One question with reference to the context. Two passages to be set from each author. Students will attempt one passage from each writer.	(15 marks)
3.	One essay-type question on each of the four writers and their texts. Students will attempt three questions.	(20 × 3 = 60 marks)

Paper-II : 18TH AND 19TH CENTURY LITERATURE

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Texts :

1.	Sheridan	:	*School for Scandal.*
2.	S.T. Coleridge	:	The following poems : *Christabel* *The Rime of the Ancient Mariner*
3.	P.B. Shelley	:	The following poems : *Ozymandias.* *Ode to the West Wind.* *To A Skylark.*
4.	George Eliot	:	*The Mill on the Floss.*

Testing :

1.	One question on age/movements with internal choice.	(15 marks)
2.	One question with reference to the context. Two passages to be set from each author at Nos. 1, 2 and 3. Students will attempt one passage from each writer.	(15 marks)
3.	One essay-type question on each of the four writers and their texts. Students will attempt three questions.	(20 × 3 = 60 marks)

संस्कृत

बी. ए. (आनर्स) द्वितीय वर्ष परीक्षा, 2011

प्रथम प्रश्न–पत्र : किसी एक नाटककार का अध्ययन

पूर्णांक : 100

समय 3 : घण्टे

उद्देश्यः

विद्यार्थियों को संस्कृत नाट्य साहित्य के प्रथम नाटक–कार भास के समय, जीवनचक्र, काव्य सौन्दर्य, प्रकृति चित्रण, प्रेम एवं सौन्दर्य–बोध, नाट्य–कला, पात्र – चरित्र चित्रण इत्यादि विषयों के विस्तृत अध्ययन के लिए प्रेरित करना ।

इसी के साथ ही 'स्वप्नवासवदत्तम्' नाटक के मूल–पाठ के अध्ययन द्वारा छात्रों में संस्कृत नाटक तथा अभिनय के प्रति रुचि उत्पन्न करना ।

विकल्प **(i)** 90+10 = 100 अंक

(अ)	भास	30 अंक
(ब)	स्वप्नवासवदत्तम्	60 अंक

नोटः– **(i)** अ' भाग के लिए 15–15 अंकों के चार भास से सम्बन्धित प्रश्न पूछे जाएं , जिनमें से दो के उत्तर देने होंगे ।

(भास का काल, काव्य,–सौन्दर्य, प्रकृति–चित्रण, प्रेम एवं सौन्दर्य, नाट्य–कला 'स्वप्नवासवदत्तम्' के पात्रों का चरित्र चित्रण)

(ii) 'ब' के लिए 10–10 अंकों की तीन सूक्तियों की सप्रसंग व्याख्या पूछी जाए, जिनमें से दो की व्याख्या करनी होगी ।

इसके अतिरिक्त 10–10 अंकों के 6 श्लोकों के अर्थ पूछे जाएं, इनमें से चार का उत्तर देना होगा ।

विकल्प (ii) : कालिदास

उद्देश्यः–

विद्यार्थियों को विश्व–प्रसिद्ध नाटककार कालिदास के काव्य, जीवनचक्र, काव्यसौन्दर्य, प्रकृति वित्रण, प्रेम एवं सौन्दर्य–बोध, नाट्य कला, पात्र चरित्र इत्यादि विषयों के विस्तृत अध्ययन के लिए प्रेरित करना ।

इसी के साथ ही 'अभिज्ञानशाकुन्तलम्' नाटक के मूल–पाठ के अध्ययन द्वारा छात्रों में संस्कृत नाटक तथा अभिनय के प्रति रुचि उत्पन्न करना ।

(अ)	कालिदास	30 अंक
(ब)	अभिज्ञान–शाकुन्तलम्	60 अंक

नोटः– (i) अ' भाग के लिए 15–15 अंक के चार कालिदास से सम्बन्धित प्रश्न पूछे जाएंगे जिनमें से दो करने होंगे ।

(कालिदास का काल, काव्य–सौन्दर्य, प्रकृति–चित्रण, प्रेम एवं सोन्दर्य–बोध नाट्य–कला, अभिज्ञान–शाकुन्तलम् के पात्रों का चरित्र चित्रण)

(ii) 'ब' भाग के लिए 10–10 अंक की तीन सूक्तियों की सप्रसंग व्याख्या पूछी जाये जिनमें से दो की व्याख्या करनी होगी ।

इसके अतिरिक्त 10–10 अंकों के 6 श्लोकों के अर्थ पूछे जाएं जिनमें से चार का उत्तर देना होगा ।

नोट – अध्यापन समय 6 घण्टे प्रति सप्ताह

द्वितीय प्रश्न–पत्र :

पूर्णांक 90+10=100

समय 3 : घण्टे

उद्देश्य :

रोचक कथा में वर्णित मानवीय जीवन के विभिन्न संघर्षो एवं अनुभवों का उल्लेख कर विद्यार्थियों को धैर्य और साहस की शिक्षा के माध्यम से संस्कृत गद्य काव्य की विशेषताओं से अवगत कराना ।

इसी के साथ ही उपनिषद् प्रतिपादित ब्रह्म के स्वरूप से परिचित कराना ।

निम्न पाठ्य पुस्तकों के उदाहरणों का अनुवाद एवं व्याख्या

1. विश्रुतचरितम्–दण्डी (6 में से 4 गद्यांश) 4×9 = 36 अंक

2. केनोपनिषद्–(6 में से 4 श्लोक) 4×9 = 36 अंक

3. उपर्युक्त पुस्तकों से सम्बद्ध अवान्तर प्रश्न पूछे जाएं ।
   (प्रत्येक पाठ्यपुस्तक में से दो–दो प्रश्न पूछे जाएंगे)
   एक–एक का उत्तर देना अनिवार्य होगा । 2×9 = 18 अंक

नोट– अध्यापन समय–6 घण्टे प्रति सप्ताह ।

URDU

B.A. (HONOURS) 2ND YEAR EXAMINATION, 2011

Courses of Reading :

100 : Marks
Time : 3 Hrs.

PAPER I : (Poetry)

1. Diwan-e-Ghalib Radif Alif. 60 Marks
2. Critical notes on Ghalib's Poetry 40 Marks

Books Recommended for Study

1. Zikr-e-Ghalib by Malik Ram
2. Afkar-e-Ghalib by Dr. Khalifa Abdur Rehman.
3. Mirza Ghalib by Dr. Amrti Lal Ishrat.

PAPER II : Essay (Literary) 100 Marks

No book is recommended

PERSIAN

B.A. (HONOURS) 2ND YEAR EXAMINATION, 2011

(Two specialized papers)

Paper : I

Marks : 100
Time : 3 Hrs.

History of Persian Literature till the end of Ghaznavi period.

Books Recommended :

1. Literary History of Persian by Dr. E.G. Brown Part I and She rul Ajam Part I and II b Maulana Shibli Nu` Mani.
2. Tareekh-e-Adabiyat-e-Iran by Dr. Raza Zadeh-e-Shafaq.

(All books can be had from Maktab-e-Jamia, Urdu Bazar, Jama Masid, Delhi-6).

Paper : II

Marks : 100
Time : 3 Hrs.

History of Persian Literature

Selijuqui period

Books Recommended

1. Literary History of Persia by Dr. E.G. Brown Part II.
2. She `ul` - Ajam by Shibli Part I, II & III.

ARABIC

B.A. (HONOURS) 2ND YEAR EXAMINATION, 2011

(Two Specialised Papers)

PAPER-I

Marks : 100
Time : 3 Hrs.

History of Pre-Islam Period

Books Recommended :

(i) History of Saracens by Ameer Ali.

(ii) Tarikh-ul-Ummat by Isalm Jai Raj Puri.

PAPER-II

Marks : 100
Time : 3 Hrs.

History of Islam, Prophet and Orthodox Caliphate.

Books Recommended :

1. Tawarekhe Islam by Prof. A.K. Fariq.
2. 'History of Sarcens' by Ameer Ali.

All books can be had from Kutub Khana-e-Rashidiyya, Urdu Bazar, Jama Masjid, Delhi-6).

FRENCH

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Paper-I : PROSE AND DRAMA

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Comprehension, explanation and literary appreciation of the prescribed textbooks.

Drama -- 45 Marks

Prose -- 45 Marks

Courses of Reading :

1. Drama-Le Bourgeois Gentilhomme—Molière (Edition Bordas).
2. La Symphonie Pastorale, Gide (Edition Bordas) ou Nouveaux Classiques Larousse.

N.B. 1. The latest syllabus will be followed and question paper should be of B.A. II (Hons.) level.

2. Choice in questions should be given.

Paper-II : APPLIED GRAMMAR & Viva-Voce

Max. Marks	:	100
Theory &Viva-Voce	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Grammar Marks : 70

1. Precis or Analysis of a given text. 15 Marks
2. Essay in French, subject or theme to be taken from the given text. 15 Marks
3. Unseen translation into French from English, Hindi, Punjabi or Urdu. 20 Marks
4. Applied Grammar. 20 Marks

Courses of Reading :

La Grammaire pour bien ècrire, Bescherelle, Hatier, 1987.

Viva-Voce : Marks : 20

(a) Dictation of an unseen passage of about 200 words based on the vocabulary and grammar covered upto B.A. II. 10 Marks

(b) Oral questions and answers on a given text (unseen). 10 Marks

N.B.

1. All unseen passages shall be based on the vocabulary and grammar covered till the end of 2nd year courses.
2. All questions (except translations) are to be asked and answered in French.
3. The latest syllabus must be given.
4. Choice in questions must be given.
5. Question paper will be set according to B.A. II (Hons.) level.

हिन्दी

बी. ए. (आनर्स) द्वितीय वर्ष परीक्षा, 2011

प्रथम पाठ्यचर्या

पूर्णांक : 90+10 = 100

समय : 3 घण्टे

कविता

1. कवितावली : (केवल सुन्दरकाण्ड) तुलसीदास
2. भारत–भारती : मैथिलीशरण गुप्त
3. लहर : जयशंकर प्रसाद

नोट :

(क) इस पाB~;चर्या में कुल मिलाकर तीनों काव्य कृतियों में से 10–10 अंकों के
छः व्याख्यांश पूछे जायेंगे। छात्रों को केवल तीन के संदर्भ, व्याख्या तथा काव्य सौन्दर्य को स्पष्ट करना होगा।

(ख) तीनों काव्यकृतियों में से कम से कम आठ समीक्षात्मक प्रश्न पूछे जाएंगे जिनमें से छात्रों को 15–15 अंकों के चार प्रश्नों के उत्तर देने होंगे।

विशेष : छात्रों को प्रत्येक कृति में से कम से कम एक प्रश्न करना अनिवार्य होगा।

(ग) **आंतरिक मूल्यांकन** अंक : 10

सहायक पुस्तकें :

1. कवितावली : सुधाकर पांडेय, लोक भारती प्रकाशन, इलाहाबाद।
2. कवितावली : गीता प्रैस, गोरखपुर।

द्वितीय पाठ्यचर्या

पूर्णांक : 90+10 = 100

समय : 3 घण्टे

1.	अंग्रेजी से हिन्दी में अनुवाद	अंक : 16
2.	निबन्ध–लेखन	अंक : 16
3.	टिप्पण तथा प्रारूपण संबंधी नियमों का समान्य ज्ञान	अंक : 18
4.	राष्ट्रीय भाषा हिन्दी, स्वरूप और महत्त्व	अंक : 10
5.	जनसंचार : रेडियो, दूरदर्शन तथा समाचार पत्रों के लिए लेखन की विशिष्ट विधि	अंक : 20
6.	हिन्दी के उपसर्ग, प्रत्यय, अर्थ व प्रयोग तथा शब्द–शोध	अंक : 10
7.	**आंतरिक मूल्यांकन**	अंक : 10

सहायक पुस्तकें :

1. टिप्पण, प्रारूपण तथा प्रूफ पठन–भोलानाथ तिवारी एवं विजय कुलश्रेष्ठ, वाणी प्रकाशन, नई दिल्ली।

2. टिप्पण, प्रारूपण तथा प्रूफ पठन–ओम प्रकाश सिंहल एवं उमापति राय चन्देल, पीताम्बर प्रकाशन, नई दिल्ली।

3. प्रचार माध्यम–राधे श्याम शर्मा, हरियाणा साहित्य अकादमी, पंचकूला, चण्डीगढ़।

4. व्यावहारिक हिन्दी–ओम प्रकाश सिंहल।

ਪੰਜਾਬੀ

ਬੀ. ਏ. (ਆਨਰਜ਼) ਭਾਗ ਦੂਜਾ (2●11) ਦੇ ਇਮਤਿਹਾਨ ਲਈ

ਪੇਪਰ ਪਹਿਲਾ: ਪੰਜਾਬੀ ਕਵਿਤਾ **ਕੁੱਲ ਅੰਕ : 1●●**

ਲਿਖਤੀ : 9●

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ : 1●

ਸਮਾਂ : 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ ਅਤੇ ਕੋਰਸ

1. ਆਸਾ ਦੀ ਵਾਰ : ਗੁਰੂ ਨਾਨਕ ਦੇਵ ਜੀ
2. ਸੱਸੀ : ਹਾਸ਼ਮ
3. ਛੇ ਰੁੱਤਾਂ : ਅੰਮ੍ਰਿਤਾ ਪ੍ਰੀਤਮ, ਆਰਸੀ ਪਬਲਿਸ਼ਰਜ਼, ਚਾਂਦਨੀ ਚੌਕ, ਦਿੱਲੀ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਤਿੰਨਾਂ ਨਿਰਧਾਰਿਤ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਤਿੰਨ ਟੋਟਿਆਂ ਦੀ ਵਿਹਾਰਕ ਆਲੋਚਨਾ, ਵਿਦਿਆਰਥੀ ਨੇ ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਇਕ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ ਕਰਨਾ ਹੈ। 1●×3 =3● ਅੰਕ

2. ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਛੋਟੇ ਸੰਖੇਪ ਪ੍ਰਸ਼ਨ (ਅੱਠ ਵਿਚੋਂ ਛੇ) ਅਤੇ ਹਰ ਪੁਸਤਕ ਵਿੱਚੋਂ ਦੋ ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣੇ ਲਾਜ਼ਮੀ ਹਨ। 6×5 =3● ਅੰਕ

3. ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਅੱਠ ਵੱਡੇ ਪ੍ਰਸ਼ਨਾਂ ਵਿੱਚੋਂ ਤਿੰਨ ਦੇ ਉੱਤਰ (ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਘੱਟੋਂ ਘੱਟ ਦੋ ਪ੍ਰਸ਼ਨ ਲਾਜ਼ਮੀ ਪੁੱਛੇ ਜਾਣਗੇ, ਬਾਕੀ ਦੋ ਪ੍ਰਸ਼ਨ ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਕਿਤੋਂ ਵੀ ਪੁੱਛੇ ਜਾ ਸਕਦੇ ਹਨ)। 3×1● =3● ਅੰਕ

ਪੇਪਰ ਦੂਜਾ : ਨਾਟਕ ਅਤੇ ਗਲਪ

ਕੁੱਲ ਅੰਕ : 1●●

ਲਿਖਤੀ : 9●

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ : 1●

ਸਮਾਂ : 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ ਅਤੇ ਕੋਰਸ

1. ਨਾਟਕ : ਰਿਸ਼ਤਿਆਂ ਦਾ ਕੀ ਰੱਖੀਏ ਨਾਂ, ਡਾ. ਆਤਮਜੀਤ, ਨਿਊ ਬੁੱਕ ਕੰਪਨੀ, ਜਲੰਧਰ

2. ਨਾਵਲ : ਕੋਠੇ ਖੜਕ ਸਿੰਘ, ਰਾਮ ਸਰੂਪ ਅਣਖੀ, ਲੋਕਗੀਤ ਪ੍ਰਕਾਸ਼ਨ, ਸਰਹੰਦ

3. ਕਹਾਣੀ ਸੰਗ੍ਰਹਿ : ਚਾਦਰ ਹੇਠਲਾ ਬੰਦਾ, ਸੁਖਵੰਤ ਕੌਰ ਮਾਨ, ਚੇਤਨਾ ਪ੍ਰਕਾਸ਼ਨ, ਪੰਜਾਬੀ ਭਵਨ, ਲੁਧਿਆਣਾ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਤਿੰਨਾਂ ਨਿਰਧਾਰਿਤ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਤਿੰਨ ਟੋਟਿਆਂ ਦੀ ਵਿਹਾਰਕ ਆਲੋਚਨਾ — 1●×3 =3● ਅੰਕ

2. ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਛੋਟੇ/ਸੰਖੇਪ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਛੇ) — 6×5 =3● ਅੰਕ

3. ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਵਿਚੋਂ ਅੱਠ ਵੱਡੇ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਤਿੰਨ ਦੇ ਉੱਤਰ (ਹਰੇਕ ਪੁਸਤਕ ਵਿਚੋਂ ਇਕ ਪ੍ਰਸ਼ਨ ਦਾ ਉੱਤਰ ਲਾਜ਼ਮੀ ਦੇਣਾ ਹੈ) — 3×1● = ਅੰਕ

GERMAN

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

PAPER-I

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment (For regular students)	:	10 Marks
Time	:	3 Hours

Note :

1. The question paper shall be set in German.
2. The mode of evaluation for Internal Assessment is to be followed as per University Guidelines.

Details of Theory Examination :

1.	Literary question on the prescribed novel.	:	30 Marks
2.	Literary question on the prescribed drama.	:	30 Marks
3.	A text for comprehension and interpretation from the textbook (Moderne Welt I)	:	30 Marks
	Total	:	**90 Marks**

Courses of Reading :

Prescribed Text Books :

1. *Der Richter und Sein Henker* by Friedrich *Dürrenmatt.*
2. *Biedermann und die Brandstiffer* by Max Frisch.
3. *Moderne Welt 1 Deutsch für Fortgeschrittene* by Heinz Griesbach (*Max Hueber Verlag München*).

Chapters :

1. *Fünftausend Sprachen.*
2. *Die demokratische Ordnung der Bundesrepublik Deutschland*
3. *Gesetzesintiative*

4. *Die Pflichten des Arbeitgebers*

5. *Die Pflichten des Arbeitnehmers*

6. *Stimullierte Bedürfnisse*

7. *Die heutige Jungend*

8. *Welche Bedeutung hat die Jugend als Alterklasse*

9. *Aus dir wird ja sowieso nichts*

10. *Ich will nicht selbständig warden*

PAPER-II : ESSAY AND TRANSLATION

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment (For regular students)	:	10 Marks
Time	:	3 Hours

Note :

1. The question paper shall be set in German.
2. The mode of evaluation for Internal Assessment is to be followed as per University Guidelines.

Details of Theory Examination:

1.	Translation of an unseen text/texts from German into English.	:	40 marks
2.	Translation of an unseen text/texts from English into German	:	20 Marks
3.	Essay writing in German (The structure of present day German Society : The family, the social role of women, youth problems, the workers, education, mass media, environment etc.)	:	30 marks
	Total	**:**	**90 marks**

DEFENCE & STRATEGIC STUDIES

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Outlines of Tests, Syllabi and Courses of Readings

Note : There will be two theory papers I and II, carrying 90 marks each. Each theory paper will have one compulsory short answer type question containing 15 questions of 2 marks each covering the entire syllabus. The candidates will be required to attempt any 10 short answer type questions. In addition, there will be four sections of the questions paper. The candidate will be required to attempt one question from each of these sections carrying 17½ marks each.

PAPER I : AREA STUDIES–PAKISTAN

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Total Teaching Periods : 75

SECTION-I

1. Military Geography of Pakistan : Its strategic importance, economic and industrial resources.
2. People and their characteristics.

SECTION-II

3. Outlines of history leading to the Birth of Pakistan.
4. Political Developments in Pakistan since 1947.
5. Role of Military in Pakistan's Politics.

SECTION-III

6. Foreign Policy of Pakistan (Basic Determinants).
7. Foreign Policy of Pakistan with regard to U.S.A. & Russia.

SECTION-IV

8. Pakistan's Foreign relations with its neighbours with particular reference to Afghanistan, China, India and Iran.

Books Recommended :

1. Akhtar, Jamna Das : Political Conspiracies in Pakistan.
2. Burke, S.M. : Pakistan's Foreign Policy.
3. Callerd, K.B. : Pakistan – A Political Study.
4. Cohen : The Pakistan Army.
5. Chopra, Surendra (Ed.) (i) : Perspectives on Pakistan's Foreign Policy.
6. Encyclopaedia Britannica.
7. Hugh, Tinker : India and Pakistan.
8. Khan, M.A. : Generals in Pakistan.
9. Khaliq, S.A. : Pakistan—Peace and War.
10. Khalid, B. Sayyad : Pakistan : Formative Phase (1857-1948).
11. Laporto Robert Jr. : Power and Privilege.
12. Moon Penomal : Divide and Quit.
13. Published by the Institute for : News Reviews on South Asia and Science and Technology, Defence Studies and Analysis, New Delhi.
14. Rushbrook, William, L.F. : The State of Pakistan.
15. Sahni, N.C. : Political Struggle in Pakistan.
16. Sidhu, K.S. : Pakistan-Khetri Adhayan (Punjabi).
17. Stephenes, Ian : Pakistan.
18. Tayyab, A. : Pakistan – A Political Geography.
19. Ziring, Lawrence : The Decisive War.
20. Ziring, Lawrence : The Ayub Khan Era.

PAPER-II : AREA STUDIES–PAKISTAN

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Total Teaching Periods : 75

SECTION-I

1. Aggression in Jammu and Kashmir Campaign 1947-48 : Causes, operations in brief and effects.

SECTION-II

2. Indo-Pak War 1965 : Causes, operations in brief and effects.

SECTION-III

3. Indo-Pak War 1971 : Causes, operations in brief and effects.
4. 1999 – Kargil War.

SECTION-IV

5. Defence Potential of Pakistan.
6. Armed Forces of Pakistan.
7. Nuclear Policy of Pakistan.

Books Recommended :

1.	Ayub Mohammad Subramanium, K.	:	Liberation War.
2.	Brines	:	Indo-Pak War (1965).
3.	Costa, Benedict	:	Dismemberment of Pakistan.
4.	Encyclopaedia Britannica.		
5.	Feldman, Herbert	:	Revolution in Pakistan.
6.	Feldman, Herbert	:	The End and the Beginning (Pakistan 1969-1971).
7.	Gupta, H.R.	:	Indo-Pak War (1965).

8. Johri, Sita Ram : Indo-Pak War (1965).

9. Kaul, B.M. : Confrontation with Pakistan.

10. Khaliq, S.A. : Pakistan—Peace and War.

11. Published by the Institute for : News Reviews on South Asia and Science and Technology, Defence Studies and Analysis, New Delhi.

12. Military Balance, Published by International Institute for Strategic Studies, London.

13. Palit, D.K. : Lightning Campaign.

14. Mankekar, D.R. : Twenty-two fateful days.

15. Mankekar, D.R. : Pakistan-cut-to-size.

16. Sahni, N.C. : Political Struggle in Pakistan.

17. Sethi, S.S. : The Decisive War.

18. Sidhu, K.S. : Pakistan-Khetri Adhyan (Punjabi).

19. Rajendra Nath (Maj. Gen.) : Military Leadership in India.

20. Sinha, S.K. : Operation Rescue.

21. Sen, L.P. (Lt. Gen.) : Slender was the Thread.

22. Ziring Lawrence : The Ayub Khan Era.

23. Lachhman Singh (Maj. Gen.) : Bangladesh War.

24. Dasgupta, C. : War and Diplomacy in Kashmir, 1947-48.

HISTORY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Note : The Honours programme outlined below is intended to encourage an indepth study of limited periods and significant epochs in Indian and World history, and wherever possible, to also provide a comparative perspective.

Any two papers from the following :

(i) Social and Cultural Trends in Ancient India.

(ii) Social and Cultural Trends in Medieval India.

(iii) Social and Cultural Trends in the Punjab (19th and 20th Century).

(i) **SOCIAL AND CULTURAL TRENDS IN ANCIENT INDIA**

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

General Instructions :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be of Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of the fifteen short answer type questions in about 25-30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall attempt ***one*** question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub-topic.

Unit-I

1. *Social Institutions :* Varna, Jati, Ashrama and Samskara.
2. *Marginalized Sections of the Society :* Shudras and Slaves.

Unit-II

3. *Women :* Status and Rights.
4. *Education :* System of learning, Principal centres.

Unit-III

5. *Early Religions :* Harappan and Vedic.
6. *Protest Movements :* Jainism and Buddhism.

Unit-IV

7. *Puranic Religions :* Vaishnavism, Shaivism and Shaktism.
8. Significant developments in art and architecture.
   (a) *Styles of Temple Architecture :* Nagara, dravida and vesara.
   (b) *Sculpture :* Mauryan, Mathura, Gandhara and Gupta.
   (c) *Painting :* Ajanta caves.

Recommended Readings :

1. P. N. Prabhu : *Hindu Social Organisation,* Popular Prakashan, Bombay, 1979.
2. R.S. Sharma : *Sudras in Ancient India,* Motilal Banarsidass, Delhi, 1980.
3. K. Roy, Ed. : *Women in Early Indian Societies,* Manohar, New Delhi, 1999.
4. A.S. Altekar : *Education in Ancient India,* The Indian Book Shop, Benaras, 1937.
5. S. Ratnagar : *Understanding Harappa,* Tulika, New Delhi, 2001.
6. A.L. Basham : *The Wonder that was India,* Rupa, Calcutta, 1992.
7. R.G. Bhandarkar : *Vaishnavism, Shaivism and Minor Religious Systems,* Indological Book House, Varanasi, 1965.
8. P. Brown : *History of Indian Architecture,* Vol. I, Taraporevala, Bombay, 1965.
9. S.K. Saraswati : *Survey of Indian Sculpture,* Munshiram Manoharlal, New Delhi, 1975.
10. D. Barrett & B. Gray : *Indian Painting,* Skira, Lausanne, 1963.

(ii) **SOCIAL AND CULTURAL TRENDS IN MEDIEVAL INDIA**

Max. Marks : 100
Theory : 90 Marks
Internal Assessment : 10 Marks
Time : 3 Hrs.

General Instructions :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be of Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of the fifteen short answer type questions in about 25-30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall attempt ***one*** question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub-topic.

Unit-I

1. *Social Conditions in the Delhi Sultanate :* Nobles, theologians, slaves and other social groups; position of women; social customs and manners.

2. *Devotional Movement :* Vaishnavism in Bengal, the Gangetic Plain and Rajputana; the teachings and impact of Kabir and Guru Nanak.

Unit-II

3. *Mystical Dimensions of Islam :* Organisation of the Chishti and Suhrawardi orders, with reference to their beliefs and practices.

4. *Landmarks in Persian Historiography :* The historical works of Ziauddin Barani and Abul Fazl.

Unit-III

5. *State Intervention in the Socio-Religious Sphere :* Akbar's liberal policy; his Ibadat Khana debates, the mahzar and breach with orthodox ulama; introduction of Tauhid-i-Ilahi and social reforms.

6. *Muslim Revivalist Movements :* Salient features of the Mahdawi movement; socio-religious ideas of Sheikh Ahmad Sirhindi and their impact.

Unit-IV

7. *Development of Literature :* The devotional writings of Tulsidas, Ravidas and Namdev; the contribution of Malik Muhammad Jaisi, Abdur Rahim Khan-i-Khanan and Ras Khan.

8. *Fine Arts in Mughal India :* The development of architecture, painting and music during the sixteenth and seventeenth centuries.

Recommended Readings :

1. K.M. Ashraf : *Life and Conditions of the People of Hindustan,* Second Edition, Munshiram Manoharlal, New Delhi, 1970.

2. Satish Chandra : *Medieval India : From Sultanate to the Mughals,* Parts I & II, Haranand Publications (Rep.), New Delhi, 2003.

3. Peter Hardy : *Historians of Medieval India : Studies in Indo-Muslim Historical Writing,* First Indian Edition, Munshiram Manoharlal, New Delhi, 1997.

4. Harbans Mukhia : *Historians and Historiography During the Reign of Akbar,* Vikas Publishing House, New Delhi, 1976.

5. Khaliq Ahmad Nizami : *Religions and Politics in India During the Thirteenth Century,* New Edition, Oxford University Press, Delhi, 2001.

6. S.A.A. Razvi : *The Wonder That Was India,* Part II, Twelfth Impression, Rupa and Company, New Delhi, 2002.

7. Mohammad Yasin : *A Social History of Islamic India,* Second Revised Edition, Munshiram Manoharlal, New Delhi, 1974.

(iii) **SOCIAL AND CULTURAL TRENDS IN THE PUNJAB (19th and 20th Century)**

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

General Instructions :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be of Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt ***nine*** out of fifteen short answer type questions in about 25-30 words each. It shall carry 18 marks and shall be **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall attempt ***one*** question from each unit – 4 in all.
4. Each essay type question will be set on half of the topics and not on a single sub-topic.

Unit-I

1. *Social Structure :* Various classes and castes with special reference to agrarian (Tenants, Zamindars and Sahukars); Commercial and Professional Classes; Village Community and Panchayat; Position of Women.
2. *Development of Literature :* Romantic and heroic ballads; the writings of Bhai Vir Singh and Socio-religious reform; the poetry of Dhani Ram Chatrik and Indian Nationalism.

Unit-II

3. *Religion :* Prevalent beliefs and practices in the 19th century; Fairs, festivals and religious grants.
4. *Trends in Art :* Development of Painting; Portraiture, Murals and Music.

Unit-III

5. *Social Awakening and Reforms :* Ahmadiyas; Arya Samaj; Singh Sabha.
6. *Contact with Christianity :* Missionary activities and response; Contribution in the field of education; Impact on Society; Influence on Education.

Unit-IV

7. *Modernization of Punjab :* Development of Education; Agriculture, Irrigation; Means of Communications.
8. *Social Impact of Partition :* Relief and Rehabilitation.

Essential Readings :

1. B.S. Saini : *The Social and Economic History of the Punjab 1901-39,* Ess Ess Publications, Delhi, 1975.
2. Bikram Jit Hasrat : *Life and Times of Ranjit Singh,* published by Author, Hoshiarpur, 1977.
3. Bhagat Singh : *Maharaja Ranjit Singh and His Times,* Sehgal Publishers Service, New Delhi, 1990.
4. Fauja Singh : *State and Society under Ranjit Singh*, New Delhi, 1982.
5. Sant Singh Sekhon : *History of Punjabi Literature,* Vol. II, Punjabi University, Patiala, 1996.
6. Kenneth W., Jones : *The New Cambridge History of India, III.1.* Religious Reform Movements in British India, Cambridge University Press.
7. John, C.B. Webster : *The Christian Community and Change in the Nineteenth Century North India,* Macmillan, 1976.
8. Satya, M. Rai : *Punjab Since Partition,* Delhi, 1986.

Reference Readings :

1. G.S. Chhabra : *Social and Economic History of Punjab*, S. Nagin & Co., Jullundur, 1962.
2. Robert D. Baird : *Religions in Modern India*, Manohar Publishers, 1991.
3. M.N. Das : *Social and Economic History of India.*
4. Indu Banga (ed.) : *Five Punjabi Centuries,* Manohar, Delhi, 2000.

POLITICAL SCIENCE

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Outlines of Test

Paper I : Option (i) : International Relations 1917-1945

OR

Option (ii) : Colonialism and Nationalism in Modern India

Paper II : Option (i) : Western Political Thought

OR

Option (ii) : Comparative Politics

Syllabus and Courses of Reading :

Paper I : OPTION (i) INTERNATIONAL RELATIONS 1917-1945

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Objectives :

The essential concern of the course is with the period between the Bolshevik Revolution and the end of the Second World War. However, to facilitate better understanding of the developments during this period, it is very essential that the teacher familiarizes the students with the major currents of international politics immediately preceding the First World War as well as during the course of that war. Needless to say, no question will be asked on the period up to the Bolshevik Revolution.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all of 18 marks each.

Course Content :

Unit-I : *The Bolshevik Revolution :*

(a) Its causes and consequences.

(b) Origins, expanse and impact of the Communist movements in the rest of the World, Communist International (Comintern).

(c) Impact of the Revolution on Soviet Foreign Policy.

Unit-II : *Nature of International Relations before and after the First World War :*

(a) Concern for peace and the efforts to ensure it through modifications in the international order.

(b) The emergence of the League of Nations.

Unit-III : 1. *Rise of Nationalism :*

(a) In Colonies.

(b) In Europe.

2. *European Politics and the Post-War Settlement :*

(a) French quest for security and the German urge for equality.

(b) Britain and the European power system.

(c) Rise of dictatorship : Fascism, Nazism and Japanese militarism.

3. The Great Depression.

Unit-IV : 1. *Origins of the Second World War, specifically :*

(a) Alliances and counter-alliances.

(b) Failure of the League of Nations.

(c) The Politics of Appeasement.

(d) American Ambivalence.

(e) Soviet-German (Molotov-Ribbontrop) Non-Aggression Pact, August, 1939.

2. The Genesis of the United Nations.

Note : Questions to be asked on the Genesis itself and not on post-1945 developments related to the United Nations.

Suggested Readings :

1. Carr, E.H. : *International Relations Between the Two World Wars (1919-1939),* (London : Macmillan Press, 1973).

2. Northedge, F.S. : *A Hundred Years of International Relations* (London : Duchworth, 1971).

3. William Shirer : *The Rise and Fall of the Third Reich, A History of Nazi Germany* (London : Secker and Warburg, 1960).

4. Keswani, K.B. : *International Relations : 1900 to Present Day* (Bombay : Himalaya Publishing House, 2004).

5. Quicky Wright : *A Study of War* (Chicago : University of Chicago Press, 1942).

6. Goodrich, L.M. and Edward Hambro : *Charter of the United Nations—Commentary and Documents* (London : London Institute of World Affairs, 1949).

7. Richard Little and Michael Smith (eds.) : *Perspectives in World Politics* (London : Routledge, 2006).

8. Chris Brown : *Understanding International Relations* (London, Palgrave, 2001).

OR

Paper I : OPTION (ii) COLONIALISM AND NATIONALISM IN MODERN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Objectives :

The paper aims at enabling the students about the nature and evolution of the Indian National Movement and constitutional development in colonial India.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a c**ompulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all of 18 marks each.

Unit-I : *Historical Foundations of National Movement :*

(a) Socio-Religious reform movements.

(b) Origin and growth of Indian National Congress.

(c) Moderates and Extremists—Debates over objectives, ideology, strategy, social basis.

Unit-II : *Gandhi and Nationalist Movement in India :*

(a) Gandhi : Political strategy and new techniques of struggle.

(b) Non-Cooperation, Civil Disobedience and Quit India Movements.

Unit-III : (a) Role of Revolutionary Nationalists, Socialist and Communist movements.

(b) Dalit, Dravidian, Women's and religion-based movements (Gurudwara Reform movement, Hindu Mahasabha, Muslim League).

Unit-IV : Landmarks in the devolution of power with special reference to Government of India : Acts of 1909, 1919, 1935 and Indian Independence Act 1947).

Suggested Readings :

1. Sumit Sarkar : *Modern India* (Madras : Macmillan, 1995).
2. Bipan Chandra : *Essays on Colonialism* (Delhi : Orient Longman, 2000).
3. Thomas Pantham et al (ed.) : *Modern Indian Political Thought* (Delhi : Sage, 1986).
4. Gail Omvedt : *Dalit Visions –The Tract of the Times* (Delhi : Orient Longman, 2002).
5. Radha Kumar : *History of Doing* (Delhi : Kali for Women, 1995).
6. Paul Brass : *The Politics of India Since Independence* (Delhi : Foundation Books, 1992).
7. V.R. Mehta and Thomas Pantham, ed. : *Political Ideas in Modern India, Thematic Exploration* (Delhi : Sage, 2006).

Paper II : OPTION (i) WESTERN POLITICAL THOUGHT

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Objectives :

This paper aims to introduce to the students the major themes of western political thought. This will be done by undertaking an indepth study of the key thinkers of this tradition.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a c**ompulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all of 18 marks each.

Unit-I	: *Plato :* The Republic	-	Justice.
		-	Philosopher-king.
		-	Ideal State.
		-	Education.
		-	Communism.
Unit-II	: *Plato :* The Statesman and the Laws	-	Second Best State.
Unit-III	: *Aristotle* : The Politics	-	Ethics and Politics.
		-	Inequality—Slavery, Women.
		-	Nature of the State.
		-	Classification of Constitutions.
		-	Revolution.
Unit-IV	: *Machiavelli* : The Prince	-	Machiavelli as a Modern Political Thinker.
		-	Morality and Politics.
		-	Power and Statecraft.

Suggested Readings :

1. Sabine, G.H. : *A History of Political Theory* (Bombay : Oxford & IBH, 1973).
2. Dunning, W.A. : *A History of Political Theories,* 3 Vols. (Allahabad : Central, 1973).
3. Sait, E.M. (ed.) : *Masters of Political Thought*, 3 Vols. (London : George and Harap, 1947).
4. William Ebenstein : *Great Political Thinkers : Plato to Present*, 1966 (New Delhi : OUP, 2000).

Reference Books :

1. Ernest Barker : *Greek Political Theory : Plato and His Predecessors* (London : Methuen, 1978).
2. Mulgan, R. : *Aristotle's Political Theory* (Oxford : Clarendon, 1977).
3. Plamenatz, John *Man and Society,* 2 Vols. (New York : Longman, Latest Edition).

4. Quentin Skinner : *Machiavelli* (Oxford : OUP, 1981).

5. Raphael, D.D. : *Hobbes : Morals and Politics* (London : George Allen, 1978).

6. Hall, J.C. : *Rousseau : Introduction to His Political Philosophy* (London : Macmillan, 1971).

7. David Maclellan : *Marx : Fontana Modern Masters Series* (London : Fontana, 1986).

OR

Paper II : OPTION (ii) COMPARATIVE POLITICS

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

Objectives :
The objective of the course is to familiarize students with recent debates and theories concerning advanced industrial as well as developing societies and this will be undertaken in a comparative framework.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks each. It shall carry 18 marks and shall be a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all of 18 marks each.

Unit-I : *Approaches to Comparative Analysis and Politics :*

(a) Traditional.

(b) Political Sociology.

(c) Political Economy.

Unit-II : *Theories of Development :*

(a) Modernisation and Development Theories.

(b) Underdevelopment and Dependency Theories.

(c) World Systems Theory.

(d) Alternative Third World perspective on development.

Unit-III : *Political Participation and Representation :*

(a) Political Parties.

(b) Interest Groups and Pressure Groups.

(c) Political Elite.

Unit-IV : (a) Theories of State : Liberal, Marxist.

(b) Theories of Advanced Capitalist States.

(c) Theories of Post-Colonial States.

(d) Civil Society and New Social Movements.

Suggested Readings :

1. Andrew Heywood : *Politics* (London : Macmillan, 1997).
2. Mary Mank Sworth and Mourice Kogan (ed.) : *The Encyclopaedia of Government and Politics* (London : Routledge, 1992).
3. Rod Hague, Martin Harrop and Shaun Breslin : *Comparative Government and Politics : An Introduction* (London : Macmillan, 1998).
4. Gabriel Almond, G. Bingham Powell, Kaare Strom and Russel J. Dalton : *Comparative Politics Today* (Delhi : Pearson Education, 2005).
5. Ronald H. Chilcote : *Theories of Comparative Politics : The Search for a Paradigm Reconsidered* (Boulder : Westview Press, 1994).

ECONOMICS

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

The candidates are required to study two papers out of the following five papers :

I Economics of Agriculture.

II Economics of Public Enterprises.

III History of Economic Thought.

IV Industrial Economics.

V Computer Applications in Economics.

Paper–I : ECONOMICS OF AGRICULTURE

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Economics of Agriculture : Its nature and scope, Interdependence between agriculture and industry. Nature and problems of agricultural development in developing countries.

Unit-II

Transforming Traditional Agriculture : Ranis & Fei, Schultz and Meller's Models, Lewis model of unlimited supplies of labour.

Unit-III

Land reforms and system of farming. Farm size and productivity relationship. Agricultural price policy, Agricultural taxation (All these topics should deal with special reference to India).

Unit-IV

New agricultural strategy and Green Revolution. Problems of small farmers and agricultural labourers. Agricultural credit, marketable and marketed surplus. Surplus and its utilization. (All these topics would deal with special reference to India).

Recommended Readings :

1. R. N. Soni : Leading Issues in Agricultural Economics, Sohan Lal Nagin Chand & Co., Jalandhar, 1992.
2. C. Eicher and L. Witt : Agriculture in Economic Development, Vera & Co., Bombay, 1970.
3. Charan D. Wadhva : Some Problems of India's Economic Policy, Tata McGraw Hills, Bombay, 1973, Part II : Articles – B, C, D, F and G.
4. A. M. Khusro : Readings in Agricultural Development, Allied, Bombay, 1968, Articles by: B. S. Minhas & T. N. Srinivasan, V. M. Dandekar, M. L. Dantwala, A. N. Khusro, T. Bergman, V. K. R. V. Rao.
5. P. C. Joshi : Land Reforms in India, Allied, Bombay, 1976, Part II, Sections I & II.
6. T. W. Schultz : Transforming Traditional Agriculture, Lyall Book Depot, Ludhiana, 1970.
7. J. W. Meller : The Economics of Agricultural Development, Vera & Co., Bombay, 1966.
8. Michael P. Todaro : Economic Development in the Third World, Orient Longman, New Delhi, 1987.
9. Francis R. Frankel : India's Green Revolution – Economic Gains and Political Costs, Oxford University Press, Bombay, 1971 .
10. S. K. Misra & V. K. Puri : Indian Economy, Himalaya Publishing House, Bombay, 1993.
11. Amarjit Singh & A. N. Sadhu : Agricultural Problems in India, Himalaya Publishing House, Bombay, 1986.

Supplementary Readings :

1. Sameeksha Trust, Bombay : Various Issues : Economic and Political Weekly.
2. Indian Society of Agricultural Economics, Bombay : Various Issues : Indian Journal of Agricultural Economics.

Paper II : ECONOMICS OF PUBLIC ENTERPRISES

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Concepts : Role of public sector in socio-economic development, Rationale of public sector—multiple objectives of public enterprises and clash and complementarity in objectives. Historical perspective of public sector in India. Organisation of public enterprises – various forms and role.

Unit-II

Role of public sector financial institutions in the financing of public enterprises. Problems of management of working capital – Management of inventory, credit and cash financing of working capital.

Unit-III

Pricing in Public Sector : Basis and types of pricing policies, pricing practices, machinery of price fixation, Role of B.I.C. & P. Accountability and profitability of public sector – Causes of poor financial performance of public sector in India.

Unit-IV

Sources of funds of public enterprises – Debt and equity, Industrial relations and its importance, Industrial disputes : causes, effects and their settlements.

Recommended Readings :

1. Laxmi Narain : Principles and Practices of Public Enterprises Management, S. Chand & Co., New Delhi, 1980.
2. -do- : Public Enterprises in India, S. Chand & Co., New Delhi, 1982.
3. Institute of Public Enterprises : Organisation in Public Enterprises, Tripathi, Bombay, 1967.
4. -do- : Pricing and Investment in Public Enterprises, Oxford and IBH Publishing Co., New Delhi.

Supplementary Readings :

1. K. R. Gupta : Issues in Public Enterprises, S. Chand & Co., New Delhi, 1978.

2. G. S. Bhalla : Financial Administration of Nationalised Industry in U.K. and India, Meenakshi, Meerut, 1968.

Paper -III : HISTORY OF ECONOMIC THOUGHT

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Pre-Classical Economic Thought : Mercantilism : Mercantilist ideas, regulations of economic life, leading theory of Mercantilist school, Physiocracy : ideas with special reference to Quesnay and Turget.

Unit-II

Classical and Marxian Political Economy : The classical tradition : Adam Smith, Ricardo, J. S. Mill and Karl Marx on value, distribution and development.

Unit-III

Reaction Against Classicism : Socialist and Nationalist thinkers, Sismondi, Saint-Simon, Robert Brown and Fourier, Subjective School : Menger, Weiser and Bohm Bewark.

Unit-IV

Neo-classicism and New Economics : Neo-classical school, time theory, Schumpeter on economic development. Keynes.

Recommended Readings :

1. Eric Roll : History of Economic Thought, Oxford University Press, Calcutta, 1974.

2. M. Blaug : Economic Theory in Retrospect, Richard D. Irwin Inc. Illinois, 1968.

3. Gide & Rist : A History of Economic Doctrine, George G. Harper & Co. Ltd., London, 1964.

4. P. M. Sweezy : The Theory of Capitalist Development, K. P. Bagchi & Co., Calcutta, 1991.

Supplementary Readings :

1. B. N. Ghosh & Rama Ghosh : Concise History of Economic Thought, Himalaya Publishing House, Delhi, 1980.

2. J.A. Schumpeter : History of Economic Analysis (Selected Portions), George Allen & Unwin, London, 1961.

3. R.R. Paul : Classical Political Economy, Kalyani Publishers, Delhi, 1979.

Paper-IV : INDUSTRIAL ECONOMICS

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Scope and methods of Industrial Economics; Basic concepts of firm, industry and market; Organisational forms and alternative motives of the firms; Industrial structure and economic development.

Unit-II

Market Structure : Sellers concentration, product differentiation, conditions of entry and economies of scale, market structure and profitability.

Growth of Firms : Vertical integration, diversification, mergers and innovation; constraints on growth-demand, financial and managerial.

Unit-III

Theory of industrial location, contribution of Weber and Sergent Florence, centralized vs. balanced regional development, industrial concentration and dispersal in India.

Unit-IV

Industrial policy since independence, Industrial development under plans and reasons for deceleration of industrial growth, Growth of State industrialization in India.

Recommended Readings :

1. R. R. Barthwal : Industrial Economics : An Introductory Text Book, New Age International Publisher (P.) Ltd., New Delhi.
2. S. C. Kuchhal : Industrial Economy of India (latest edition), Allahabad, Chaitanya Publishing House.
3. P. J. Devine et al. : An Introduction of Industrial Economics, George Allen & Unwin Ltd., London, 1974.
4. W. Steward Howe : Industrial Economics : An Applied Approach, The Macmillan Press Ltd.
5. Rudar Datt & K. P. M. Sundharam : Indian Economy (latest edition), S. Chand & Company.

Supplementary Readings :

1. David M. Smith : Industrial Location : An Economic Geographical Analysis, John Wiley & Sons Inc.
2. E. T. Penrose : The Theory of Growth of Firm, Oxford, Basil Blackwell.
3. Harry W. Richardson : Elements of Regional Economics, Penguin Education.
4. Ishar Judge Ahluwalia : Industrial Growth in India : Stagnation Since Mid Sixties, Oxford University Press, Delhi.
5. Sivayya & Das : Indian Industrial Economy, S. Chand & Co., New Delhi.

Paper- V : COMPUTER APPLICATIONS IN ECONOMICS

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Total Teaching Hours	:	60

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Introduction to Computers : Simple Model of a Computer; Steps in problem solving on a computer; Data representation; Input/Output devices; Computer memory; Central Processing Unit (CPU); Computer generations and classifications.

Unit-II

Introductory Programming : Algorithm, Computer Program and Programming languages; Basic data types and declarations; Elementary PASCAL programming using READ and WRITE statements.

Unit-III

Programming : Control structures (Iterative and conditions), Structured data type (Array only); Introduction to functions and procedures.

Unit-IV

Programming Applications : Development of algorithm and programs for simple statistical and mathematical methods like computation of measures of central tendency; dispersion, acceleration, simple regression, matrix operations (Addition, substraction, multiplication and inverse).

Recommended Readings :

1. V. Rajaraman : Fundamentals of Computers, PHI, Prentice-Hall India, Delhi, 1991.
2. V. Rajaraman : Computer Programming in PASCAL, PHI, Delhi, 1990.

Supplementary Readings :

1. G. M. Schnaider and Others : An Introduction to Programming and Problem Solving with PASCAL, PHI, Delhi, 1984.
2. E. Balaguruswami : Computer Oriented Statistical and Numerical Methods, Macmillan, Delhi, 1988.

SOCIOLOGY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

A student opting for B.A. (Hons.) in Sociology will be required to take up 2 papers in the 2nd year and 2 papers in the 3rd year out of the following :

Paper I : (SOC 251) : Social Thought (Compulsory) (Written : 90 marks)
(Internal assessment : 10 marks)

Paper II : Any one of the two **Options :**

Option (i) (SOC 252) Social Psychology (Written : 90 marks)
(Internal assessment : 10 marks)

Option (ii) (SOC 253) Modern Indian Social Thought (Written : 90 marks)
(Internal assessment : 10 marks)

Paper-I (SOC 251) : SOCIAL THOUGHT (Compulsory)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Note :
1. For written paper, the students will be required to attempt **five** questions in all. Question No. 1 will be **compulsory** and cover the entire syllabus. In addition to it, students will attempt **four** essay type questions, one from each of the four units.
2. Question No. 1 will be compulsory comprising of short answer type questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).
3. On an average, 15 hours are to be devoted to each unit.
4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

This paper intends to familiarize the students with the contributions of major thinkers in classical Sociology and their continuing relevance to its contemporary concerns. This paper offers the Honours students with a better idea of the development of Sociology as a discipline, compared with those in the General Course.

Course Content :

UNIT-I

Auguste Comte : Laws of Three Stages, Hierarchy of Sciences, Positivism.

UNIT-II

Emile Durkheim : Social facts and division of labour, social solidarity; suicide and religion.

UNIT-III

Max Weber : Social action, class, status and power, types of authority; Protestant ethic and the spirit of capitalism.

UNIT-IV

Karl Marx : Historical materialism; dialectical materialism; class conflict, capitalism and theory of social change.

Suggested Readings :

1. Coser, L.A. : *Masters of Sociological Thought*, New York : Harcourt Brace Jovonovich, 1997.
2. Collins, Randall : *Theoretical Sociology,* Jaipur : Rawat Publications, 1997.
3. Raymond, Aron : *Main Currents in Sociological Thought, Vols. I & II*, Harmondsworth: Penguin Books, 1982.
4. Timasheff, N.S. : *Sociological Theory : Its Nature and Growth*, New York : Random House, 1967.
5. Zeitlin, I.M. : *Ideology and the Development of Sociological Theory*, New Jersey : Prentice Hall, 1987.

Further Readings :

1. Harlambos, M. : *Sociology, Themes and Perspectives*, Delhi : Oxford University Press, 1984.
2. Lemert, C. (ed.) : *Social Theory – The Multicultural & Classical Readings,* Jaipur : Rawat Publications, 2004.
3. Marx & Engles : *The Communist Manifesto*, Harmondsworth : Penguin Books, 1967.
4. Morrison, Ken, Marx, Durkheim, Weber : *Formation of Modern Social Thought*, London : Sage, 1995.
5. Sorokin, P. : *Contemporary Sociological Theories*, Ludhiana : Kalyani Publishers, 1978.

Paper-II : Opt (i) (SOC 252) : SOCIAL PSYCHOLOGY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Note :

1. For written paper, the students will be required to attempt **five** questions in all. Question No. 1 will be **compulsory** and cover the entire syllabus. In addition to it, students will attempt **four** essay type questions, one from each of the four units.

2. Question No. 1 will be compulsory comprising of short answer type questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).

3. On an average, 15 hours are to be devoted to each unit.

4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

This paper exposes the honours students in Sociology to the related field - Social Psychology. The purpose of this paper is to introduce the basic concepts in Social Psychology, which have a bearing on the Social-Psychological understanding of social interaction, social groups, public opinion and leadership qualities. This course prepares the students to study social phenomena from social-psychological perspectives.

Course Content :

UNIT-I

Introduction – Nature, Scope, Application and Relationship of Social Psychology with Sociology.

UNIT-II

Perception, Motivation, Learning, Attitudes and their Measurement.

UNIT-III

Social Groups – Groups and Classification. Collective Behaviour : Crowd, Audience, Mob and their differences.

UNIT-IV

Propaganda and Public Opinion.
Leadership – Emergence, type and functions of leaders, leadership qualities.

Essential Readings :

1. Curtis : *Social Psychology*, New York : McGraw Hill, 1960.
2. Fraser, C. & Burchell, B. : *Introducing Social Psychology,* Cambridge : Polity Press, 2001.
3. Kuppuswamy, B. : *Social Psychology*, Bombay, Asia Pub. House, 1980.
4. Krech, D. & Cretchfield, Richards : *Theory & Problems of Social Psychology*, New York : McGraw Hill, 1948.
5. Lindgren, H. Clay : *An Introduction to Social Psychology,* 2nd ed., New Delhi : Wiley Eastern, 1974.
6. Linde Smith, Strauss, A.R., Denzin, A.L. & Norman, D. : *Social Psychology,* New Jersey : Prentice Hall, 1988.
7. Perry, J.B. & Pugh, M.D. : *Collective Behaviour : Response to Social Stress,* Minnisotta : West Publishing Co., 1978.

Paper-II : Option (ii) (SOC 253) : MODERN INDIAN SOCIAL THOUGHT

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Note :
1. For written paper, the students will be required to attempt **five** questions in all. Question No. 1 will be **compulsory** and cover the entire syllabus. In addition to it, students will attempt **four** essay type questions, one from each of the four units.
2. Question No. 1 will be compulsory comprising of short answer type questions. It will consist of 12 short answer type questions of 2 marks each, out of which the students are required to attempt 9 short answer type questions ($9 \times 2 = 18$ marks).
3. On an average, 15 hours are to be devoted to each unit.
4. Question Nos. II to IX will be long answer questions, two questions from each unit with internal choice, consisting of 18 marks each ($4 \times 18 = 72$ marks).

Objective :

The basic purpose of this paper is to introduce the students with the contribution of scholars to the development of Modern Indian Social Thought from different perspectives.

Course Content :

UNIT-I

Indological/Textual Approach : G.S. Ghurye, P.H. Prabhu, Louis Dumont.

UNIT-II

Liberal Reformist Approach : Raja Ram Mohan Roy, B.R. Ambedkar, Jawaharlal Nehru.

UNIT-III

Revivalist Approach : Swami Dayanand, Aurobindo Ghosh, Swami Vivekananda.

UNIT-IV

Gandhian Approach : Social, Economic and Political Thought of Mahatma Gandhi.

Essential Readings :

1. Ghosh, Aurobindo : *Foundation of Indian Culture*, Pondicherry : Sri Aurobindo Ashrama,1959.
2. Bandhopadhyaya, Jayantanuja : *Social and Political Thought of Gandhi*, Bombay : Allied Publishers, 1969.
3. Haimsath, Charles, H. : *Indian Nationalism and Hindu Social Reforms*, New Jersey : Princeton University Press, 1964.
4. Momin, A.R. : *The Legacy of G.S. Ghurye : A Centennial Festschrift*, Bombay : Popular Prakashan, 1996.
5. Nag, Kalidas (ed.) : *The Works of Raja Ram Mohan Roy, Vols. I & II,* Calcutta : Sadharana Bramo Samaj, 1962.
6. Reddy, P.S., and Gangadhar, V. (eds.) : *Indian Society; Continuity, Change and Development,* New Delhi : Commonwealth Publishers, 2004.
7. Singh, Yogendra : *Modernization of Indian Tradition*, Delhi : Thomson Press, 1993.

Further Readings :

1. Ghosh, Aurobindo : *Essay on Gita*, Pondicherry : Sri Aurobindo Ashrama, 1955.
2. Bose, N.K. (ed.) : *Selection from Gandhi*, Ahmedabad : Navjiwan Press, 1955.
3. Bose, N.K. : *Studies in Gandhism*, Ahmedabad : Navjiwan Publishers, 1957.
4. Prabhu, P.H. : *Hindu Social Organization,* Bombay : Popular Prakashan, 1998 (reprint).
5. Singh, Yogendra : *Indian Sociology, Social Conditioning and Emerging Concerns,* New Delhi : Vistaar Publications, 1986.

PHILOSOPHY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Paper-I : CONTEMPORARY WESTERN PHILOSOPHY & CONTEMPORARY INDIAN THOUGHT AND CULTURE

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Aims and Objectives :

The aim of this paper is to acquaint students with the theories of different western contemporary philosophers and the problems of language, perception, meaning and truth, science and God; and to familiarise the students with Indian thought and culture according to Contemporary Indian thinkers.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Course Contents :

Unit-I

1. Refutation of Idealism.
2. Realism.
3. Common Sense and Perception.
4. Nature of Logical Necessity—Internal & External Relations.

Unit-II

1. Language, Meaning & Truth.
2. Philosophy, Science.
3. God.
4. Existentialism : Theistic and Atheistic Approaches.

Unit-III

Rabindra Nath Tagore : Mahatma Gandhi & Tilaka.

Unit-IV

1. Sri Aurobindo
2. Swami Dayanand Saraswati, Swami Vivekananda.

Essential Readings :

1. Inquiry into Meaning & Truth by Russell, Harper Collins, 1980.

2. Philosophy in the 20th Century by A.J. Ayer, London, Unwin, 1982.

3. Modern Indian Thought by V.S. Narvane, Asia, 1967.

Further Readings :

4. Religion of Man by Rabindranath Tagore.

5. Srimad Bhagwad Gita Rahasya or Karma Yoga Sastra by B.G. Tilak, Manohar Book Pub., 2002, Delhi.

6. Contemporary Philosophy : Studies of Logical Positivism and Existentialism by F.C. Copleston, Continuum Intl. Pub. Group, 2003.

Paper II : STUDY OF A SPECIAL PHILOSOPHER

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Aims and Objectives :
The aim of this paper is to make students study a philosopher indepth amongst the options offered.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

One out of the following Philosophers :

Option (i) : *PLATO*

Unit-I : *Virtues :* Wisdom, Courage, Temperance and Justice, Justice and Happiness : Is the just life happier than the unjust one ?

Unit-II : The meaning and purpose of education.

Unit-III : Nature of knowledge and its relation to sensation.

Unit-IV : Language and its relation to the world : Do names picture the objects named ?

Essential Readings :

1. Republic.
2. Theatetus.
3. Cratylus in Plato's Dialogues tr. B. Jowett.

Suggested Readings :

1. *The Philosophy of Plato* by Rupert C. Lodge, Routledge & Kegan Paul, London, 1956.
2. *PLATO* by J.C.B. Gosling, Routledge & Kegan Paul, London, 1973.
3. *A Critical History of Greek Philosophy* by W.T. Stace, Macmillan India Ltd., 1982.

OR

Option (ii) : *ARISTOTLE*

Unit-I : Form and Matter.

Unit-II : Aristotle's Ten Categories.

Unit-III : Four Causes and Change.

Unit-IV : Identity of Thought and Truth in the Active Reason.

OR

Option (iii) : *IMMANUEL KANT*

Unit-I : A –priori and A –posteriori judgements.

Unit-II : Nature of Space and Time.

Unit-III : Function of Categories and Significance of Antinomies.

Unit-IV : Refutation on the various proofs for the existence of God.

OR

Option (iv) : *UPANISHADS*

(Chandogya, Mandukya and Isa)

Unit-I : Mandukya's Treatment of 'Aum'. Symbol of Immanent and Transcendent aspects of Atman.

Unit-II : Philosophy of Being as instructed by Uddalaka Aruni in Chandogya Upanishad.

Unit-III : Nature of Atman in Isa Upanishad.

Unit-IV : Views of Upanishads on Brahman, Soul and World.

OR

Option (v) : *SARTRE* : Being and Nothingness

Unit-I : Being-in-itself and Being-for-itself.

Unit-II : Facticity.

Unit-III : Freedom, responsibility and badfaith.

Unit-IV : Being for Others, Existence of Others, the problem of embodiment.

Essential Readings :

1. *Being & Nothingness* by J.P. Sartre, Washington Square Press, 1993.
2. *Cambridge Companion to Sartre ed.* by Christina Howells, Cambridge University Press, 1992.

OR

Option (vi) : *NAGARJUNA*

Unit-I : Concept of Sunya.

Unit-II : Sunyata and Pratiyasamutpada.

Unit-III : Logic and Reality.

Unit-IV : Dialectical System.

OR

Option (vii) : *SAMKARA*

Unit-I : Theory of Causation.

Unit-II : Prakriti as a Ultimate Cause.

Unit-III : Parinamavada Vs. Vivartavada.

Unit-IV : Refutation of Prakritikaranavada.

OR

Option (viii) : ***PATANJALI***

Unit-I : Language and meaning.

Unit-II : Word-meaning relationship.

Unit-III : Purpose of grammar.

Unit-IV : Sabda as a Pramana.

Essential Readings :

1. Plato-Republic, Theatetus, Cratylus In : B. Jowett (Tr.) Plato's Dialogues (New York : Random, 1937).

2. Aristotle-Metaphysics Ed. & Tr. by John Washington and with an introduction by Sir David Ross (London : Dentsons, 1956).

3. Kant-Critique of Pure Reason : Norman Kemp Smith (Tr.) (London, Macmillan, 1952).

4. Upanisads : Chandogya, Mandukya, Isa.

5. Sartre—Being and Nothingness.

6. Nagarjuna-Madhyamika Karika, Stcherbatsky Theo : Conception of Buddhist Nirvana (Delhi : Bharatiya Vidya Prakashan).

7. Samkara, Brahmasutrabhasya : Tarka Pada (First four Sutras, Thibaut, Sacred Books of the East, Delhi, Motilal Banarasi Dass, 1956).

8. Patanjali : Mahabhasya (Chaps. I & II Gita Press, Gorakhpur).

PSYCHOLOGY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Paper I : SOCIAL PSYCHOLOGY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

I Social Psychology course emphasizes the relationship between society and individuals. It also enables the students in understanding group dynamics, role of attitudes and leadership behaviour in the society.

II Pedagogy of the Course Work :

80 % Lectures (including expert lectures).
20 % assignments, discussion and seminars.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

There shall be 9 questions in all. Five questions are required to be attempted. The first question shall be Short Answer type containing 12 short questions spread over the whole syllabus, and the students are required to attempt 9 short answer type questions in about 25 to 30 words each. Each short question will carry 2 marks, totalling 18 marks, and shall be a c**ompulsory** question. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit –4 in all.

Unit I : *Introduction :* Nature of Social Psychology, Methods of Research in Social Psychology.

Unit II : *Attitudes :* Concepts, Link between Attitude and Behaviour, Formation of Attitudes, Attitude Change, Measurement of Attitudes.

Unit III : *Groups :* Nature and Functioning, Group Effects on Task Performance.

Unit IV : *Leadership :* Emergence, Functions of Leaders, Theories of Leadership (Trait Theories, Behavioural Theories, Fiedler's Contingency Model).

Books Recommended :

Essential Books :

1. Baron, R.A. and Byrne, D. (2003) : Social Psychology : Understanding Human Interaction, New Delhi : Prentice Hall of India.

2. Krech, D., Crutchfield, R. S. and Ballachey, H. (1979) : Individual in Society, New York : McGraw Hill.

3. Luthans, F. (1998) : Organizational Behaviour, New York : Irwin McGraw Hill.

Reference Books :

1. Lindgren, H.C. (1973) : Introduction of Social Psychology, New York : Wiley.

2. Robbins, S.P. (1998) : Organizational Behaviour, New York : Prentice Hall of India.

Paper-II : SYSTEMS OF PSYCHOLOGY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

I This course aims at helping the students to appreciate the basic assumptions, principles and historical roots of modern scientific psychology. It will also help them to understand main theoretical perspective in psychology.

II Pedagogy of the Course Work :

80 % Lectures (including expert lectures).
20 % assignments, discussion and seminars.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

There shall be 9 questions in all. Five questions are required to be attempted. The first question shall be Short Answer type containing 12 short questions spread over the whole syllabus, and the students are required to attempt 9 short answer type questions in about 25 to 30 words each. Each short question will carry 2 marks, totalling 18 marks, and shall be a c**ompulsory** question. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit –4 in all.

Unit-I	:	*Associationism*	:	Basic Postulates; Older Associationists, Thorndike; Pavlov; Guthrie.
		Structuralism	:	Basic Postulates; Weber and Fechner; Wundt; Titchener.
Unit-II	:	*Functionalism*	:	Basic Postulates; James; Dewey and Angell; Carr; Woodworth.
		Behaviourism	:	Basic Postulates; Watson; Skinner; Hull; Tolman; Bandura.

Unit-III : *Gestalt :* Basic Postulates; Wertheimer, Kohler and Koffka; Lewin.

Psychoanalysis : Basic Postulates; Freud; Adler; Jung.

Unit-IV : *Neo-Psychoanalysis :* Basic Postulates; Horney; Sullivan; Erikson.

Humanistic Existential : Basic Postulates, Maslow; Rogers; Existential Philosophers.

Books Recommended :

Essential Books :

1. Marx, M.H., and Cronan-Hillix, W.A. (1987) : Systems and Theories in Psychology, New York : McGraw Hill.
2. Chaplin, J.P., and Kraweic, T.S. (1984) : Systems and Theories of Psychology, New York : Holt, Rinehart, and Winston.
3. Woodworth, R.S., and Sheehan, M.R. (1964) : Contemporary Schools of Psychology, London : Methuen.

Reference Books :

1. Hall, C.S., Lindzey, G. and Campbell, J.B. (1998) : Theories of Personality, New York : John Wiley and Sons.
2. Bower, G.H. and Hilgard, E.R. (1981) : Theories of Learning, New Delhi, Prentice Hall of India.
3. Wolman, B.B. (1979) : Contemporary Theories and Systems in Psychology, Delhi : Freeman.

GEOGRAPHY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Paper-I : THE NATURE OF GEOGRAPHY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

1. To introduce the students to the philosophical and methodological foundations of the subject and its place in the world of knowledge.
2. To familiarise them with the major landmarks in development of geographic thought at different periods of time.

Course Content :

UNIT-I

The basic objective of geography; relevance of geography to societal problems; the role of a geographer in planning.

Contribution of Eratosthenes, Ptolemy, Humboldt and Ritter.

UNIT-II

Empirical/theoretical approach in geography.

Geography as the study of : (i) distributions, (ii) man-environment relationship, (iii) areal differentiation, and (iv) spatial organisation.

UNIT-III

Environmental determinism versus possibilism.

Relationship between geography and history; dualism between physical and human geography, dualism between regional and systematic geography.

UNIT-IV

Time and genesis in geography.

Behavioural geography.

Quantitative revolution in geography.

Note : 1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).

2. The whole syllabus will be divided into 4 units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at Serial Number 1.

3. Internal assessment may include written assignments, snap tests, participation in class discussions, term papers, attendance etc.

Books Recommended :

Essential Readings :

1. Husain, Majid. : *Evolution of Geographical Thought,* Rawat, Jaipur, 2001.

2. Martin, G.J. : *All Possible Worlds – A History of Geographical Ideas*, Oxford University Press, U.K., 2005.

Further Readings :

1. Adhikari, Sudeepta : *Fundamentals of Geographic Thought,* Chaitanya, Allahabad, 2004.

2. Dikshit, R.D. : *Geographical Thought : A Contextual History of Ideas*, Prentice Hall, New Delhi, 2000.

3. Hartshorne, Richard : *Perspectives on the Nature of Geography,* Rand, McNally and Co., Chicago, 1959.

4. Minshull, R. : *The Changing Nature of Geography,* Hutchinson University Library, London, 1970.

Pedagogy :

- Students of geography may be encouraged to interact with their counterparts from other disciplines and discuss the nature of the subject.
- The students may be encouraged to collect information on any theme amenable to geographical interpretation.

Paper-II : ***Any one of the following optional courses :***

Option (i) : POPULATION GEOGRAPHY (An Elementary Course)

OR

Option (ii) : URBAN GEOGRAPHY (An Elementary Course)

Option (i) : POPULATION GEOGRAPHY (An Elementary Course)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

The course is meant to provide an understanding of spatial and structural dimensions of population and the emerging issues. The course is further aimed at familiarising the students with global and regional level problems and also equip them for comprehending the Indian situation.

Course Content :

UNIT-I

Nature and scope of population geography, factors affecting distribution of population; present world pattern of population distribution. Density of population – various types of densities and their significance; broad spatial patterns of density of population in the world.

UNIT-II

Growth of population – Determinants of population growth; trends of population growth in developed and developing countries of the world.

Migration – Types, causes and consequences.

UNIT-III

Sex composition – Its determinants and world patterns.

Occupational structure – A comparative study of developed and developing countries. Literacy–its determinants and world patterns.

UNIT-IV

Urbanization – Definition, process of urbanization; comparative study of urbanization in developed and developing countries.

Note : 1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).

2. The whole syllabus will be divided into 4 units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at Serial Number 1.

3. Special credit will be given to suitable use of maps.

4. Internal assessment may include written assignments, snap tests, participation in discussions in the class, term papers, attendance etc.

5. For reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) I their case shall be of Maximum Marks allotted to the paper(s) concerned.

Books Recommended :

Essential Readings :

1. Chandna, R.C. : *A Geography of Population*, Kalyani Publishers, Delhi, 1996.

2. Clarke, J.I. : *Population Geography,* Pergamon Press, Oxford, Latest Edition.

Further Readings :

1. Chandna, R.C. : *Jansankhya Bhoogol*, Kalyani Publishers, New Delhi, 1996.

2. Crook, N. : *Principles of Population and Development*, Pergamon, New York, 1997.

3. Jones, H.R. : *Population Geography*, Harper and Row, London, 1981.

4. Sundaram, K.V. & Nangia, S. (Eds.) : *Population Geography*, Heritage, New Delhi, 1986.

5. Trewartha, G.T. : *The Less Developed Realm – A Population Geography*, McGraw-Hill, New York, 1972.

6. Trewartha, G.T. : *The More Developed Realm : A Geography of its Population,* Pergamon Press, Oxford, 1979.

Pedagogy :

1. Use maps and atlases.
2. Students should be encouraged to read Population Clock.
3. Consult census publications.
4. Organised field work.
5. Organise discussions on population issues.

Option (ii) : URBAN GEOGRAPHY (An Elementary Course)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

1. To understand the process of urbanization and origin, growth and classification of urban settlements with relevant theories and models.
2. To examine the changing economic base and structure of the contemporary cities.
3. To relate urbanisation process and the evolution of the urban system.
4. To examine the contemporary urban issues and suggest new urban planning and urban policy.

Course Content :

UNIT-I

Definition, nature and scope of urban geography.

Definition of urban places.

Evolution of City through Historical Times : Ancient, medieval, modern.

UNIT-II

Site and situation of cities : Definition, importance, types.

Size and spacing of cities.

Ecological Processes : Centripetal and centrifugal forces, types of ecological processes.

UNIT-III

Internal structure of cities : Theories of internal structure and their evaluation, concentric zone theory, sector theory, multiple nuclei theory.

UNIT-IV

City – Region relations.

Umland of Cities : Definition, bases of demarcation, attributes.

Rural –Urban Fringe : Attributes and demarcation.

Note :

1. The compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts in about 25-30 words each. Each part will carry 2 marks (total 20 marks).
2. The whole syllabus will be divided into 4 units. Eights questions will be set out of the whole syllabus, 2 from each Unit. The students will be required to attempt one question from each unit and each question will carry 17.5 marks. These will be in addition to the compulsory question at Serial Number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion, in the class, term papers, attendance etc.
4. For reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

Books Recommended :

Essential Readings :

1. Carter, Harold : *The Study of Urban Geography*, Edward Arnold, London, Latest Edition.
2. Krishan, Gopal : *Nagar Bhugol,* Punjab State University Text Book Board, Chandigarh, 1974.
3. Mayer, H.M. and Kohn, L.F. (Eds.) : *Readings in Urban Geography*, University of Chicago Press, Chicago, 1964.

Further Readings :

1. Johnson, H.J. : *Urban Geography : An Introductory Analysis,* Pergamon Press, London, 1967.
2. Knox, Paul : *Urban Social Geography : An Introduction,* Longman, London, 1982.

Pedagogy :

- Awareness to data sources such as from Census of India, Town Planning organizations, and their application to understand and evaluate the spatial patterns and the processes of urbanisation should be highlighted in the class. This needs to be in the form of selected case studies.
- Study of urban morphology and urban functions with special reference to selected towns need to be encouraged.
- Atlases and maps of NATMO and Census should be consulted and students should be given opportunity to participate in group discussions.

PUBLIC ADMINISTRATION

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Outline of Tests, Syllabi and Courses of Reading

Paper- I : ADMINISTRATIVE THOUGHT

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : 1. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

2. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (2) in question paper in this regard.

Critical Study of the following writers :

Unit-I

Kautilya, M.K. Gandhi and Woodrow Wilson.

Unit-II

Frederick W. Taylor, Henri Fayol, Luther Gulick, and L.B. Urwick.

Unit-III

Max Weber, Elton Mayo, M.P. Follet.

Unit-IV

Herbert A. Simon, Abraham Maslow and Fred W. Riggs.

Essential Readings :

1. Bertram, M. Gross : The Managing of Organisations : The Administrative Struggle (London : MacMillan Publishers, 1964).

2. George, C.S. : History of Management Thought (New Delhi : Prentice Hall, 1974).

3. Fred W. Riggs : Administration in Developing Countries (Boston Houghton, 1964).

4. Herbert A. Simon : Administrative Behaviour : A Study of Decision Making Process in Administrative Organisation (New York : Macmillan, 3rd ed., 1976).

5. Sapru, R.K. : Theories of Administration (New Delhi : S. Chand, 1996).

6. Maheshwari, S.R. : Administrative Thinkers (New Delhi : MacMillan, 2003).

Further Readings :

1. Pugh, D.S. (ed.) : Organisation Theory (Harmondsworth : Penguin Books, 2nd Ed., 1984).

2. David Silverman : The Theory of Organisation (London : Heinemann, 1970).

3. Chester Barnard : The Functions of the Executive (London : Oxford University Press, 1974).

4. Jack Rabin & Others : Handbook of Public Administration, 2nd Edition (New York : Marcel Dekker, 1977).

PAPER-II : The candidates have to select one out of the following three options :

Option (i) : Agricultural Administration in India

Option (ii) : Industrial Development Administration in India

Option (iii) : Police Administration

Paper- II Option (i) : AGRICULTURAL ADMINISTRATION IN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : 1. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

2. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (2) in question paper in this regard.

Unit-I

Role of agriculture in economic development.

Problems of agricultural development.

Agriculture Policy.

Unit-II

Agriculture Development under Five-Year Plans.

Centre-State relations in Agriculture.

Agriculture Administration at the Central and State Levels.

Unit-III

New technology for the development of agriculture.

Marketing of agriculture produce.

Agriculture finance and rural indebtedness.

Unit-IV

Rural Development Programmes.

Agriculture education, research and extension.

World Trade Organisation and Agriculture.

Essential Readings :

1. Ghosh, R.N. : Agriculture in Economic Development, with special reference to Punjab (New Delhi : Vikas, 1977).

2. Chaudhary, P. : Readings in Indian Agricultural Development (London : George Allen & Unwin, 1972).

3. Mahajan, V.S. : Agriculture and Rural Economy in India (New Delhi : Deep & Deep Publications, 1990).

4. Hanumantha Rao, C.H. : Technological Change and Distribution of Gains in Indian Agriculture (Delhi : Macmillan, 1975).

Further Readings :

1. N.C.A.E.R. : Credit Recruitment for Agriculture (New Delhi : 1974).
2. Government of India, Commission of Agriculture : Reports (New Delhi : Manager of National Publications, 1976).
3. Government of India, Ministry of Finance, Economic Division : Economic Survey (2009-10).
4. Government of India, Planning Commission : 10th Five Year Plan 2002–2007 (New Delhi : Planning Commission, 2002).

OR

Paper-II Options (ii) : INDUSTRIAL DEVELOPMENT ADMINISTRATION IN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : 1. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

2. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (2) in question paper in this regard.

Unit-I

Rationale and problems of industrialization.

Role of industry in economic development.

Unit-II

Industrial Policy Resolutions of 1948, 1956 and New Industrial Policy, 1991.

Industrial Development and Regulation Act, 1951 (as Amended).

Unit-III

Industrial development under Five Year Plans.

Public Enterprises and Industrial Development.

Industrial Dispersal Policy and Problems.

Unit-IV

Memorandum of Understanding.

Privatization of Public Enterprises.

Essential Readings :

1. Kuchhal, S.C. : Industrial Economy of India (Allahabad : Chaitnaya Publishing House, 1983).
2. Marathe, S.S. : Regulation and Development : The Indian Policy Experience of Controls over Industry (New Delhi : Sage Publications, 1986).
3. Rudar Dutt : Indian Economy (New Delhi : S. Chand, 2009).
4. Aggarwal, A.N. : Indian Economy (New Delhi : Vikas Publishing House, 2001).

Further Readings :

1. Sivayya, K.V. and Dass, V.B.M. : Indian Industrial Economy (New Delhi : S. Chand & Co., 2004).
2. Govt. of India, Planning Commission : 10th Five Year Plan 2002–2007 (New Delhi : Planning Commission, 2002).
3. Dhar, P.K. : Indian Economy–Its Growing Dimensions (New Delhi : Kalyani Publishers, 2007).
4. Government of India, Ministry of Finance, Economic Division : Economic Survey (2009-10).

PAPER-II – Option (iii) : POLICE ADMINISTRATION

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : 1. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidates would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one question from each unit. Each question will carry 18 marks.

2. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (2) in question paper in this regard.

Unit-I

Need and Importance of Police in Society – Evolution of Police System in India – the Landmarks since 1861. Police Reforms in India.

Unit-II

The Police and the Union Government – Ministry of Home Affairs and the State Policing System. The District Police in India _ District Police Organization and its Working.
Organisation of Village Police Force.

Unit-III

Police Personnel Administration in India : The IPS and State Police Services :

Recruitment, Training, Promotion and Service conditions; Discipline in Police Services.

Unit-IV

Public Expectations from Police and their implications on the role of Police. The Concept and Need of Community Policing. Concept & Importance of E-Governance in Police. Need and Means of Attitudinal Change in Police.

Essential Readings :

1.	Gupta, Anandswarup	:	The Police in British India 1861-1947, Bureau of Police Research and Development, New Delhi, 2007.
2.	Chaturvedi, J.C.	:	Police Administration and Investigation of Crime, Isha Books, Delhi, 2006.
3.	Ghosh, S.K. & Rustamji, K.F.M.	:	Encyclopaedia of Police in India, Vol. I, II III, 1993, 1994 and 1997.
4.	Vdackumchery, James	:	National Police Commission : Issues for Rethinking, APH Publishing Corporation, New Delhi, 1998.
5.	O. Glenn Stahl	:	Public Personnel Administration, Oxford & IBH Publishing Co., New Delhi, 1975.
6.	Muthuswamy, P.	:	Swamy's Manual on Disciplinary Proceedings, Swamy Publishers (P.) Ltd., Chennai (1989).

ANCIENT INDIAN HISTORY, CULTURE & ARCHAEOLOGY

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Note : **The students shall have to opt for two papers for 2nd year examination out of the following four papers :**

OUTLINES OF TEST

Paper I	:	Social ideas and social life from the earliest times to 1000 A.D.
Paper-II	:	Economic ideas & economic life from the earliest times to 1000 A.D.
Paper-III	:	Political ideas and political life from the earliest times to 1000 A.D.
Paper-IV	:	Spread of Indian Culture in Central Asia, China and Tibet.

Paper-I : SOCIAL IDEAS AND SOCIAL LIFE FROM THE EARLIEST TIMES TO 1000 A.D.

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :

1. The primary objective of this paper is to acquaint the students with the origin and development of social institutions and ideas so peculiar to Indian society. It will also made a survey of the social life of the people in ancient India.

2. **Pedagogy of the Course Work :**
The students should be given an idea of the chronology and contents of the sources mentioned above. The changing pattern in social life should also be highlighted.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions. It shall carry 18 marks i.e. 2 marks of each and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

1. Sources of social ideas and life : Vedic Literature, Epics, Manu Smriti and Kamsutra of Vatsayayana.
2. Four Purusharthas; Sixteen Samsakaras.

Unit-II

3. Origin and development of caste system.
4. Slavery in ancient India.
5. Stages of life—four asramas.

Unit-III

6. Family life, law of inheritance.
7. Development of education in Ancient India.
8. Dress, ornaments, foods and drinks.

Unit-IV

9. Position of women in ancient India.
10. Institution of marriage.

Paper-II : ECONOMIC IDEAS & ECONOMIC LIFE FROM THE EARLIEST TIMES TO 1000 A. D.

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :

1. The main objective of this paper is to familiarize the students with economic ideas as contained in the secular and non-secular literature of our country. It will also make a survey of the economic life in ancient India to help the students to understand economic problems of the people.

2. **Pedagogy of the Course Work :**
The students should be given an idea of the chronology, authorship and contents of the sources mentioned above. The changing condition of agriculture, industries, and trade and commerce should also be highlighted.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions. It shall carry 18 marks i.e. 2 marks of each and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

1. Sources of Economic Life : Vedic literature, Epics, Arthasastra of Kautilya and Nitisara of Kamandaka.

2. Agriculture in ancient India, ownership of land.

Unit-II

3. Taxation.

4. Industries and industrial products in ancient India.

Unit-III

5. Types of labour, wages.

6. Introduction and Development of coinage.

Unit-IV

7. Trade and Commerce in ancient India.

8. Guild System.

9. Trade routes and transport.

Paper-III : POLITICAL IDEAS AND POLITICAL LIFE FROM THE EARLIEST TIMES TO 1000 A.D.

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :

1. The emphasis of this paper will be on the development of political ideas as enshrined in our secular and non-secular literature. It will also make a study of the administrative systems during various periods focusing on the ideologies of the kings.
2. **Pedagogy of the Course Work :**
   The students are expected to know the broad rules laid down by the lawgivers and have knowledge of the administrative organization in different periods. They should be familiarized with the changing patterns of administrative set up at different times.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions. It shall carry 18 marks i.e. 2 marks of each and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

1. *Sources of Political Ideas and Life :* Vedic literature, Epics, the Arthasastra of Kautilya, Smritis of Manu and Yajnavalkya.

2. *Political Institutions :* State, King, Council of Ministers, Foreign Relations.

Unit-II

3. Administrative organization during the vedic period.
4. Administration of the Mauryas.

Unit-III

5. Administration of Kushanas.
6. Administration of the Guptas.

Unit-IV

7. Administration of Harsha.
8. Administration of the Cholas.

Paper IV : SPREAD OF INDIAN CULTURE IN CENTRAL ASIA, CHINA AND TIBET

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :

1. The primary aim of this paper is to make the students study the spread of Indian culture and ideas in Central Asia, China and Tibet, generally termed in history as 'Greater India'. This forms an important phase of India's contribution to the world culture.

2. **Pedagogy of the Course Work :**
   It is expected to acquaint the students with the background of Indian culture's spread and the factors responsible for it. While outlining the spread of Indian culture and religion, the spread of art, architecture and literature should also be taken care of. Visual aids wherever required should be used to familiarize students with Indian art and its influence in these countries.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions. It shall carry 18 marks i.e. 2 marks of each and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Routes and nomadic movements in Central Asia.

Unit-II

Spread of Indian Culture in Tokharistan, Sogdiana and the Pamir states.

Unit-III

Introduction of Buddhism in China and the early contacts.

Unit-IV

Missionaries and Pilgrims :

(a) Indian Missionaries to China from Central Asia, Kashmir, Central and Western India.

(b) Chinese pilgrims in India— Fa–Hien, Hsuan Tsang and I-Tsing.

(c) Spread of Buddhism in Tibet up to 112th century A.D. with special reference to the missions of Padma Sambhava and Aisa.

(d) Lamaism and Tibet Buddhist Literature.

MUSIC (VOCAL AND INSTRUMENTAL)

B.A. (HONOURS) SECOND YEAR EXAMINATION, 2011

Note : *General Instructions :*

1. Harmonium will not be allowed as an accompaniment in Vocal Music. The candidate has an option to take either Vocal or Instrumental Music as a subject. He/She has the option to take any one of the following instruments :
   Sitar, Sarangi, Veena, Sarod, Dilruba, Violin, Bansuri, Shahnai and Tabla.
2. Either Bhatkhande or Vishnudigamber notation system may be followed.
3. The theory paper shall be of three hours duration and practical paper shall be of 45 minutes duration.

Paper I : (Theory)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Note : There will be 9 questions in all and the question paper will be divided into 5 units. The students will attempt 5 questions choosing at least one question from units I to IV and Unit Vth of the question paper which is based on unit V of the syllabus will be compulsory and that will be of 22 marks. The remaining 4 questions will of 17 marks each.

UNIT-I

1. Concept and characteristics of Raga Dhyan.
2. Raga-Ragini system of classification of Ragas.
3. Detailed study of Dhrupad and Khayal styles.

UNIT-II

1. Brief biographical sketches and contributions towards music of the following Musicians :
   (a) Ustad Mushtaq Ali Khan.
   (b) Ustad Alladiya Khan.
   (c) Ustad Imdad Khan.
   (d) Ustad Karim Khan.
2. Detailed Study of Ghranas of the following :
   (a) Khyalshailley.
   (b) Sitar Vadan.

UNIT-III

1. A study of the following :
   Thumri, Masitkhani, Razakhani and Sitarkhani Gats, Dadra, Tappa.
2. Study of Tanas, its varieties and their importance in North Indian Music.

UNIT-IV

1. Brief knowledge of the following :
   Nada, Swara, Shruti, Nyas, Vadi, Laya, Tala, Zamzama, Jod Alap, Gat, Present Raglakshanas, Avirbhava and Tirobhava.

2. Comparative study of Bhatkhande and Vishnu Digamber Notation System.

UNIT-V

1. Description and Comparative study of Detailed and Non-detailed Ragas and to write the notations of Detailed ragas.
   (a) Detailed Ragas :
   Puriya, Darbari, Hamir Shankra.
   (b) Non-detailed Ragas :
   Adana, Sohni, Kamod, Bahar.

Paper–II : Practical (Choice & Viva) 90 Marks

Internal Assessment : 10 Marks
Time : 45 Minutes

(a)	Choice (Solo performance)	:	45 Marks
(b)	Practical Test	:	45 Marks
	(i) Tuning	:	10 Marks
	(ii) Capacity to play Talas on Tabla	:	05 Marks
	(iii) Capacity to play Harmonium	:	05 Marks
	(iv) Viva	:	25 Marks

Part-A : (CHOICE OF THE CANDIDATE)

(a) A candidate has to demonstrate in the Raga of his/her choice from the prescribed ragas with detailed elaborations :

(i)	A Vilambit Khayal	:	15 marks
(ii)	A Drut Khayal	:	10 marks

OR

(i)	A Vilambit Gat	:	15 marks
(ii)	A Drut Gat	:	10 marks

(b) One Dhrupad with different Layakaries in any detailed ragas 10 marks

OR

One Vilambit Gat in *Chautala* with different layakaries 10 marks

(c) Tarana or one Fast Gat in Ektal 05 marks

(d) Any one of the following : 05 marks
Bhav-Geet, Bhajan, Shabad or Dhun

Part-B : (VIVA)

(a) List of Detailed Ragas :
- (i) Darbari.
- (ii) Hamir.
- (iii) Shankra.
- (iv) Puriya.

(b) Non-detailed Ragas, description with Aroh, Avroh and pakad in the following Ragas :
- (i) Adana.
- (ii) Sohni.
- (iii) Kamod.
- (iv) Bahar.

(c) Talas : Dadra, Kaharva, Jhaptal, Ektal and Teental on Tabla.

(i) Two slow khayals with extempore Alaps and Tanas in any of the detailed Ragas. 15 marks

OR

Two slow Gatas with extempore Alaps and Toras in any of the detailed Ragas.

(ii) One fast khayal or one fast gat with elaboration in each of the prescribed Ragas. 10 marks

(iii) Ability to recite Dhamar, Adachautal, Deepchandi, Jhoomra, and Punjabi talas in single. 5 marks.

(iv) Tuning of your instrument. 5 marks

(v) Capacity to play on Tabla the Talas prescribed in the course. 5 marks

(vi) Capacity to play on the Harmonium any raga from the choice. 5 marks

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HOME SCIENCE

The syllabus of Home Science for B.A. (Honours)/B.Sc. (Honours) second year has been **KEPT IN ABEYANCE.**

PHYSICS

B.Sc. (HONOURS) SECOND YEAR EXAMINATION, 2011

Note : There will be two papers of 100 marks each.

I. Theoretical Physics
II. Atomic Spectroscopy and Lasers

Each paper will consist of 3 sections :

Section –A	:	4 questions
Section – B	:	4 questions
Section – C	:	1 question

Contents of sections A and B are indicated in the syllabi for respective papers. Sections C will consist of one question carrying 10 parts of short answer type, covering the whole syllabus. The candidate will attempt the compulsory question and two questions each from sections A and B, five questions in all. All questions will carry equal marks.

Paper I : Theoretical Physics

Marks : 100
Time : 3 Hours

SECTION A : Mathematical Physics

I Differential Equations :

Some partial differential equations in physics, Methods of separation of variables; separation of Helmholtz equation in cartesian, spherical and cylindrical coordinates. Laplace equation in various coordinate systems, choice of coordinate system. Ordinary and singular points; series solution around an ordinary and a regular singular points; getting a second solution – Wronskian method and alternative method. (2.1-2.3, 3.1-3.5 of Book 1).

II Boundary Value Problems :

Boundary value problems in physics, boundary value problems and series solution, examples; Eigen values, eigen functions and Sturm-Liouville problem, Hermitian operators and their eigen values & eigen functions. (4.1-4.5 of Book 1).

III Some Special Functions in Physics :

Legendre polynomials, associated Legendre functions and spherical harmonics, Hermite polynomials, Laguerre and associated Laguerre polynomials. The gamma function. The Dirac delta function. (5.4-5.9 of Book 1).

SECTION B : Quantum Mechanics

I Linear Vector Spaces :

Definition of a linear vector space, linear independence, basis and dimension. Scalar product, orthonormal basis, Gram-Schmidt orthogonalization, linear operators, matrices, orthogonal, Unitary and Hermitian matrices, Eigenvalues and eigenvectors of a matrix, Matrix

representation of linear operators, eigenvalues and eigenvectors of operators, Hermitian and Unitary operators, Simultaneous eigenvectors and commutativity, Infinite dimensional spaces, Hilbert space, linear vector spaces and basic postulates of quantum mechanics, Parity operator, Fundamental commutation relation, Heisenberg's equation of motion, commutation rules and uncertainty principle. (7.1-7.15 of Book 1, 6.1-6.12, 9.1-9.10 of Book 2).

II Scattering theory :

Introduction, wavefuntions for a free particle in spherical polar coordinates. Expansion of plane wave in spherical harmonics, scattering by a short-range field. Method of partial waves, dependence of δ_l on l and E., potential well. Low energy scattering, scattering by an attractive square potential well. (8.1-8.8 of Book 2).

Books :

1. Chattopadhyay, P.K. : Mathematical Physics (Wiley Eastern, New Delhi, 1990).
2. Powell, J.L. & Crasemann, B. : Quantum Mechnics (Narosa & Co., New Delhi, 1981).

Paper II – Atomic Spectroscopy and Lasers

Marks : 100
Time : 3 Hours

SECTION A : Atomic Spectra

I Vector Atom Model :

The spinning electron, space quantization, quantum numbers and their physical interpretation. Quantum numbers of an atom, magnetic moments of an atom and Lande 'g' factor, Larmor's theorem, Stern-Garlach experiment, Spectral terms and their notations–Positronium.

II Pauli Exclusion Principle :

Exclusion principle, maximum number of electrons in a given group or sub-group, ground state of an atom, equivalent electrons, multiplicity and Pauli's principle.

III Spectra of Alkali Atoms :

Different series in alkali spectra, Ritz combination principle, term values in alkali spectra and quantum defect, spin-orbit interaction, doublet structure in alkali spectra, transition rules, intensity rules, alkali like spectra, coupling schemes, interaction energy in LS-coupling, interaction energy in jj-coupling, selection rules for atoms of two valence electrons, singlet and triplet series in two valence electrons, two electron atom-He atom, spectrum of He atom, spectrum of mercury.

IV Fine Structure of Spectral Lines :

Basic facts and Sommerfeld theory, fine structure of Hydrogen lines, qualitative brief of Pauli Darwin theory and quantum mechanical treatment for fine structure, Lamb Shift.

V Complex Spectra :

Terms in many electron atom, term in equivalent electron atom system, Hund's rule, Lande interval rule, normal and inverted terms, order of the fine structure levels, selection rules in many electron atom, energy level diagram for complex atom, regularities in atomic spectra of complex atoms.

SECTION-B : Laser

I Introduction :
Directionality, intensity, monochromaticity, coherence.

II Einstein's Quantum Theory of Radiation :
Einstein's Coeffs., momentum transfer, lifetime, possibility of amplification.

III Masers :
Two level ammonia maser, hydrogen maser, three-level maser system.

IV Simple Optical Processes :
Wave and interference, coherence, coherence of the field and the size of the source, coherence and monochromaticity, kinematics of optical absorption.

V Basic Principles of Lasers :
Population inversion, laser pumping (two and three level systems), resonators, vibrational modes of a resonator, number of modes per unit volume. Open resonators, confocal resonator, quality factor, losses inside the cavity, the threshold condition, quantum yield.

VI Types of Lasers :
Ruby laser, its pumping power & spiking. He-Ne laser, CO_2 laser.

VII Dynamics of Laser Processes :

Q-switching, mechanical shutter, electro-optical shutters, shutters using saturable dyes, peak power emitted during the pulse. Giant pulse dynamics, laser amplifiers, mode locking.

VIII Applications :
Holography : Its principle, recording & reconstruction of the image, theory and practical applications (Qualitative).

Non-linear Optics : Harmonics generation, second harmonic generation, phase matching, third harmonic generation, optical mixing.

Books :

1. Gupta, S.L., Kumar, V. and Sharma, R.C. : Elements of Spectroscopy (Pragati Prakashan, Meerut).
2. Laud, B.B. : Lasers and Non-linear Optics (Wiley Eastern Limited, New Delhi, 2nd Ed.)

CHEMISTRY

B.Sc. (HONOURS) SECOND YEAR EXAMINATION, 2011
Outlines of Tests

			Lecture Hours	*Marks*	
				Theory	*Internal Assessment*
Paper I	:	Organic Chemistry	40	90	10
Paper II	:	Analytical Chemistry	40	90	10

Paper I : ORGANIC CHEMISTRY

Max. Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time	:	3 Hours
Lectures	:	40

OBJECTIVE OF THE COURSE :

To teach the fundamental concept of Chemistry and their applications, the syllabus pertaining to B.Sc. (HONOURS) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

Unit-I

1. **Rearrangements and Neighbouring Group Effects :** (10 Lectures)
Hoffmann, Curtius, Lossen-similarities, advantages and stereochemistry at the migrating group, effect of migrating group on rate of reaction. Baeyer Villiger Rearrangement, Hydroperoxide rearrangement–migratory aptitude, Brown, Pinacol, Wolff, Stevens, Wittig and Favorskii rearrangements-participation of groups such as –O–C–R, halogens, Aryl, and sigma bond, nonclassical carbocation.

Unit-II

2. **Stereochemistry :** (10 Lectures)
Enantiomeric relationship, Diastereomeric relationship, Prochiral relationship. Conformational relationship in carbocyclic rings other than 5-membered, conformational effects on reactivity. Relationships between ring size and facility of ring closure, Hard Soft Acid Base principles: pka Values. Generation of Carbon Nucleophiles by Deprotonation, Regio selectivity and stereo selectivity in enolate formation, Medium effects, O versus C as site of alkylation, examples of Stork enamine reactions.

Unit-III

3. **Organic Chemistry of Life :** (10 Lectures)

Nucleic Acids : Nucleosides; Pentoses, heterocyclic bases, nucleotide; Purines and phosphate, DNA and RNA Hydrogen bonding and DNA replication (heredity), enzymes, nomenclature of enzymes, characteristics of enzymes, coenzyme A and prosthetic group. Coenzyme derived from niacin and examples of redox reaction with it.

Unit-IV

4. **Pericyclic Reactions :** (10 Lectures)

Electrocyclic Reactions: Woodward and Hoffmann rules, correlation diagram for cyclobutene = butadiene interconversion, Frontier orbital theory. Examples of 4, 6 and 8 electron systems, stereochemical modes.

Cycloaddition Reactions : Woodward and Hoffmann rules, Frontier Molecular orbital theory for Dieis-Alder reaction, (4+2) and (2+2) cycloadditions, exo and endo isomers, Examples Bicyclo (2, 2, 1) heptane and Bicyclo (2, 2, 2,) octane systems. Intramolecular Diels Alder reaction.

Sigmatropic Rearrangements: Woodward and Hoffmann rules, Frontier Molecular orbital for 1, 3; 1, 5 and 1, 7 shifts. Examples including vitamin D_2 formation from 7-dehydro cholesterol (Previtamin D_3).

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Suggested Books :

1. Morrison, R.T., Boyd, R.N., Organic Chemistry; 6th edition, Pubs: Prentice Hall India Pvt. Ltd., 1992.

2. Carey, F.A., Sundberg, R.J., Advanced Organic Chemistry : 4th edition, 2001.

3. Mukherji, S.M., Singh, S.P., Organic Chemistry; 1st edition, Pubs: New Age International, New Delhi, 1985, Vol. 2.

Paper II : ANALYTICAL CHEMISTRY :

Max. Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time	:	3 Hours
Lectures	:	40

OBJECTIVE OF THE COURSE :

To teach the fundamental concept of Chemistry and their applications, the syllabus pertaining to B.Sc. (HONOURS) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

Unit-I

1. **Elementary Concepts :** (10 Lectures)

Qualitative and quantitative analysis, concepts and importance of quantitative analysis, choice of methods for analysis, sampling and theories of sampling, preparation of samples for analysis, calibration standards, solution concentrations in terms of various conventions, simple equilibrium calculations, calibration of analytical weights and glass wares, significance of calibration, role of instrumental methods of analysis in analytical chemistry.

Terms of mean, medium, precision and accuracy in chemical analysis, types of errors in analysis, determining accuracy of methods, improving accuracy of analysis, data treatment for series involving relatively few measurements, least square curve fitting, propagation of errors, significant figure convention, standard deviations, confidence limits.

Theory of volumetric and gravimetric methods of analysis, primary and secondary standards, theory of indicators and their selection for titrations, precipitation methods, optimum conditions for precipitation, steps involved in gravimetric analysis, important organic precipitants.

Unit-II

2. **Acid-Base Equilibria :** (5 Lectures)

Acids and bases, ionization equilibria for weak monobasic acids, bases and related salts, acid base titration reagents and indicators for neutralization titrations, neutralization titrations for a strong acid with a strong base, polybasic acid with a strong base.

3. **Oxidation-Reduction Equilibria :** (5 Lectures)

Oxidation-reduction reactions, reduction potentials, Nernst equation, oxidation-reduction titrations and curves, oxidation-reduction indicators, applications of oxidation-reduction techniques, determination of water by the Karl Fischer titration method.

Unit-III

4. **Precipitation Equilibria :** (5 Lectures)

Solubility of precipitates, effect of competing equilibria on solubility of precipitates, separation of ions by control of concentration of precipitating reagents, effect of electrolyte concentration on solubility product, theory of precipitation titrations. The Vohlard and the Mohr's methods of analysis, adsorption indicators.

5. **Complexation Equilibria :** (5 Lectures)

Complexation, formation constants, EDTA equilibria, effect of pH on EDTA equilibria, complexometric titration curves, selection of indicators for complexometric titrations, applications of complexometric equilibria.

Unit-IV

6. **Solvent Extraction and Ion-Exchange Separation :** (6 Lectures)

Basic concepts of solvent extraction, relationship between distribution coefficient, distribution ratio and per cent extraction, applications of extraction methods in analytical chemistry, ion-exchange processes, ion-exchange-resins, techniques and applications of ion-exchange separation, basic concepts of column and gas chromatography.

7. **Radioisotope Methodology and its Applications :** (4 Lectures)

Radioactive isotopes, radioactive decay rates and half-life periods of radioactive isotopes, sample preparation in radiochemical measurements, background, dead-time and inefficiency corrections in radioactive measurements, isotope dilution method and its applications, radiometric-titrations, neutron activation analysis.

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Suggested Books:

1. Skoog, D.A., West, D.M., Saunders, H., Fundamentals of Analytical Chemistry, 8th edition, International, 2004.

2. Khopkar, S.M., Basic Concepts of Analytical Chemistry, 18th edition, Pubs, New Age International, 2002.

BOTANY

B.Sc. (HONOURS) SECOND YEAR EXAMINATION, 2011

		Theory	*Int. Assess.*	*Total*	*Practical*	*Int. Assess.*	*Total*
Theory Paper-A :	Plant Physiology and Ecology	68	+ 7	75	22.5	+ 2.5	25
Theory Paper-B :	Development of Plants and their Utilization	68	+ 7	75	22.5	+ 2.5	25
	Total :			**150**			**50**

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory papers A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Paper -A : PLANT PHYSIOLOGY AND ECOLOGY

Max. Marks : 75
Theory : 68 Marks
Internal Assessment : 7 Marks

Objective :

The main objective of this paper is to make the students aware about various functions and metabolisms of plants. The course material of this paper would enable the students to understand the role of environment in causing structural and functional variations in plants. In fac this paper would provide the students an insight into eco-physiology – an upcoming field dealing with interaction between plant growth metabolism and environment.

Teaching Methodology :

Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

PLANT PHYSIOLOGY

UNIT-I

PLANT PHYSIOLOGY

Plant-Water Relations :
Water transport processes; diffusion and osmosis; water potential and chemical potential; absorption of water, water transport through tracheids and xylem; transpiration and its significance; factors affecting transpiration; mechanism of stomatal movement.

Mineral Nutrition :
Criteria of essentiality of elements; macro-and micronutrients; role of essential elements; mineral deficiency symptoms and plant disorders; nutrient uptake and transport mechanisms; role of cell membrane, ion pumps and carriers.

Photosynthesis :
Historical background and significance; structure of photosynthetic apparatus; photosynthetic pigments; accessory pigments and the photoprotective carotenoids; reaction center complexes; photochemical reactions; electron transport pathways in chloroplast membranes; photophosphorylation; the Calvin cycle; the C_4 carbon cycle; crassulacean acid metabolism; synthesis of starch and sucrose; photorespiration.

Transport of Organic Substances :
Transport of photosynthates; source-sink relationships; the mechanism of translocation in the phloem; assimilate partitioning.

UNIT-II

Respiration :
Glycolysis; the TCA cycle and its regulation; electron transport in mitochondria; oxidative phosphorylation; pentose phosphate pathway; cyanide-resistant respiration.

Nitrogen Metabolism :
Biological nitrogen fixation; reduction of N_2 into ammonia; *nif* genes; regulation of nitrate reductase and nitrogenase; nitrate and ammonium assimilation.

Growth and Development : General aspects – definitions, phases of growth; kinetics of growth; physiology of seed dormancy and seed germination; concept of photoperiodism; Physiology of flowering; the florigen concept and role of hormones; vernalization; senescence and fruit ripening; importance of respiratory climacteric; discovery, physiological role and mechanism of action of the phytohormones – auxins, cytokinins, gibberellins, abscisic acid and ethylene; photomorphogenesis; discovery of phytochromes and cryptochromes, their role and mechanism of actions; signal transduction – basic concept; plant movements – tropic and nastic; biological clocks.

UNIT-III

ECOLOGY

Introduction :
Holocoenotic nature of environment; limiting factors; ecological amplitude triggering factors (soil, water, atmosphere).

Ecological Adaptations :
Concept, ecads, ecotypes and ecoclines; adaptations in relation to soil oligotrophy; adaptive significance of photosynthetic pathways.

Population :
Concepts; density and pattern; idealized plant life history; population growth; carrying capacity; population regulation; r and k selection; population interactions.

Community :
Community characteristics and their measurement, species diversity (alpha, beta and gamma), niche.

Ecosystem :
Concept, components and organisation, energy flow; ecological efficiencies; cycling of C, N and P; characterization and structure of ecosystem; the biotic and abiotic components, their inter-relationship; the processes within ecosystem, the trophic organisation; autotrophy, heterotrophy, parasitism, detritus, decomposition.

Ecological Succession :
Mechanism and types; nature of climax; food chains and food webs; ecological pyramids.

UNIT- IV

Flow of Energy and Materials :
Flow of energy and materials within ecosystem; models of energy flow; ecosystem productivity; biogeochemical cycles; major types of ecosystems.

Plant Indicators and their role in Environmental Monitoring.

Phytogeography :
General principles, vegetation of India.

Suggested Readings :

1. Barbou, M.G., Burk, J.H. and Pitts, W.D., 1987 : *Terrestrial Plant Ecology, Benjamin/Cummings,* Publications Co., California.

2. Galston, A.W., 1989 : *Life Processes in Plants*, Scientific American Library, Springer-Verlag, New York, USA.

3. Hill, M.K., 1997 : *Understanding Environmental Pollution,* Cambridge University Press.

4. Hooykaas, P.J.J., Hall, M.A. and Libbenga, K.R. (eds.), 1999 : *Biochemistry and Molecular Biology of Plant Hormones, Elsevier,* Amsterdam, The Netherlands.

5. Hopkins, W.G., 1995 : *Introduction to Plant Physiology*, John Wiley & Sons, Inc., New York, USA.

6. Kormondy, E.J., 1996 : *Concepts of Ecology,* Prentice-Hall of India Pvt. Ltd., New Delhi.

7. Mackenzie, A. et al., 1999 : *Instant Notes in Ecology*, Viva Books Pvt. Ltd., New Delhi.

8. Moore, T.C., 1989 : *Biochemistry and Physiology of Plant Hormones (2nd edition), Springer-Verlag, New York, U.S.A.*

9. Odum, E.P., 1983 : *Basic Ecology*, Saunders, Philadelphia.

10. Salisbury, F.B. and Ross, C.W., 1992 : *Plant Physiology* (4th edition), Wadsworth Publishing Co., California, U.S.A.

11. Taiz, L. and Zeiger, E., 1998 : *Plant Physiology* (2nd edition), Sinauer Associates, Inc., Publishers, Massachusetts, U.S.A.

Suggested Laboratory Exercises

Teachers may collect plants/materials available in their locality/institutions

Max. Marks : 25
Practical : 22.5 marks
Internal Assessment : 2.5 Marks

Plant Physiology

1. To study the permeability of plasma membrane using different concentrations of organic solvents.
2. To study the effect of temperature on permeability of plasma membrane.

3. To extract enzymes and study their activity, e.g. amylase, lipase, acid phosphatase, catalase, peroxidase.

4. Effect of the following factors on enzyme activity :

   pH, temperature, enzyme and substrate concentration, heavy metals.

5. Comparative study of rate of respiration of various plant parts.

6. To extract and separate chloroplast pigments by solvent method and demonstrate fluorescence in chloroplast extracts.

7. To determine the osmotic potential of vascular sap by plasmolytic method.

8. To determine the water potential of given tissue (any tuber).

9. Bioassay of plant hormones – auxins, ethylene, GA, ABA and cytokinin.

10. To determine stomatal index, stomatal frequency and percentage of leaf area open through stomata. Also to study the effect of ABA on stomatal closure.

11. Role of light in germination of photoblastic seeds e.g. *Lactuca sativa, Arabidopsis.*

Suggested Readings (for laboratory exercises)

1. Copleand, R.A., 1996 : *Enzymes : A Practical Introduction to Structure, Mechanism and Data Analysis,* VCH Publishers, New York.

2. Dryer, R.L. and Lata, G.F., 1989 : *Experimental Biochemistry*, Oxford University Press, New York.

3. Moore, T.C., 1974 : *Research Experiences in Plant Physiology : A Laboratory Manual,* Springer-Verlag, Berlin.

4. Wilson, K. and Goulding, K.H. (Eds.), 1986 : *A Biologist's Guide to Principles and Techniques of Practical Bio-chemistry,* Edward Arnold, London, UK.

Suggested laboratory exercises

ECOLOGY

1. To determine the working and use of instruments for the measurement of temperature (soil, air, water), moisture (rainfall, relative humidity, soil moisture), wind (velocity and direction) and light intensity.

2. To study selected soil properties by spot test : Texture, pH, carbonate, nitrate, base deficiency and reductivity.

3. To study ecological adaptations (morphological and anatomical) in plants (hydrophytes, xerophytes, epiphytes).

4. To determine minimum area of sampling unit (quadrat) for the study of grassland community.

5. To estimate Importance Value Index for grassland species.

6. To study leaf form spectrum of a woodland.

7. To prepare maps of India with respect to (i) major climatic zones, (ii) forest types, and (iii) biogeographical regions and to comment on it.

Suggested Readings (for laboratory exercises)

1. Misra, R., 1968 : *Ecology Work Book,* Oxford & IBH, New Delhi.

2. Moore, P.W. and Chapman, S.B., 1986. : *Methods in Plant Ecology,* Blackwell Scientific Publications.

3. Smith, R.K., 1996 : *Ecology and Field Biology,* Harper Collins, New York.

Paper -B : DEVELOPMENT OF PLANTS AND THEIR UTILIZATION

Max. Marks : 75
Theory : 68 Marks
Internal Assessment : 7 Marks

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory papers A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Objective :

This paper deals with internal structure of various plant parts, their growth patterns and structural abnormalities along with utilization of plants. This paper would provide the students an idea about importance of various plant parts as well as plant wealth such as medicinal, crop and food plants; fibre and oil yielding plants, timber yielding plants etc.

Teaching Methodology :

Teaching methodology includes series of lectures, making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

UNIT-I

Organization of the Higher Plant Body :
The shoot and root systems; variation in habit and longevity; environmental influences.

Meristems and Development : Shoot apical meristem, root apical meristem, lateral meristems and their functions.

Range of form and structure of stem, leaf and root; their tissues and functions.

UNIT-II

Secondary Body of the Plant :
Vascular cambium; secondary xylem (basic structure of wood); secondary phloem and periderm.

Role of Wild Plants in Ecosystem Functioning :
A general account.

Domestication of Plants :
Primary and secondary centers of diversity; plant introduction.

UNIT-III

A General account of wheat, rice, maize, sorghum, bajra, potato and sugarcane.

Legumes :
Chickpea (Bengal gram), red gram (arhar) and fodder legumes – a general account.

Vegetable Oil Sources :
Mustard, groundnut, soyabean and coconut –a brief account.`

Plant Fibres :
Cotton, jute and coir.

Timber and Firewood Species :
A general account of use of properties of *any ten* plants of your region.

UNIT-IV

Medicinal Plants :
A brief account of *ten* important plant drugs and their chief constituents used in indigenous and allopathic systems of medicine.

Natural Rubber (Hevea brasiliensis), essential oils, insecticides and dyes – a concise account.

Ornamental Plants :
Familiarity with seasonals and perennials grown in your locality

Suggested Readings :

1. Carlquist, S., 1988 : *Comparative Wood Anatomy : Systematic, Ecological and Evolutionary Aspects of Dicotyledonous Wood,* Springer-Verlag, Berlin.
2. Cutter, E.G., 1969 : *Cells and Tissues, Part I.,* Edward Arnold, London.
3. Cutter, E.G., 1971 : *Plant Anatomy : Experiment and Interpretation, Part II, Organs,* Edward Arnold, London.
4. Esau, K., 1977 : *Anatomy of Seed Plants,* 2nd edition, John Wiley and Sons, New York.
5. Fahn, A., 1974 : *Plant Anatomy,* 2nd edition, Pergamon Press, Oxford.
6. Fuller, K.W. and Gallon, J.R., 1985 : *Plant Products and New Technology,* Clarendon Press, Oxford, New York.
7. Kocchar, S.L., 1988 : *Economic Botany in Tropics,* 2nd edition, Macmillan India Ltd., New Delhi.
8. Lyndon, R.F., 1990 : *Plant Development : The Cellular Basis,* Unwin Hyman, London.
9. Mauseth, J.D., 1998 : *Plant Anatomy,* The Benjamin/Cummings Publishing Company Inc., Menlo Park, California, USA.
10. Nair, M.N.B., 1988 : *Wood Anatomy and Major Uses of Wood,* Faculty of Forestry, Universiti Putra Malaysia, 43400 Serdang, Selangor D.E., Malaysia.
11. Raghvan, V., 2000 : *Developmental Biology of Flowering Plants,* Springer-Verlag, New York.
12. Raven, P.H., Evert, R.F. and Eichhorn, S.E., 1999. : *Biology of Plants,* 5th edition, W.H. Freeman and Co., Worth Publishers, New York.

13. Sambamurthy, A.V.S.S. and Subramanyam, N.S., 1989 : *A Textbook of Economic Botany,* Wiley Eastern Ltd., New Delhi.

14. Sharma, O.P., 1996 : *Hill's Economic Botany,* Tata McGraw Hill Publishing Company Ltd., New Delhi.

15. Simpson, B.B. and Conner-Ogorzaly, M., 1986 : *Economic Botany –Plants in Our World,* McGraw Hill, New York.

16. Steeves, T.A. and Sussex, I.M., 1989 : *Patterns in Plant Development,* 2nd edition, Cambridge University Press, Cambridge.

17. Thoms, P., 2000 : *Trees : Their Natural History,* Cambridge University Press, Cambridge .

18. Tippo, O. and Stern, W.L., 1977 : *Humanistic Botany,* W.W. Norton, New York.

19. Vishnu Swarup, 1997 : *Ornamental Horticulture,* Macmillan India Ltd., New Delhi.

Suggested Laboratory Exercises

Teachers may collect plants/materials available in their locality/institutions

Max. Marks	:	25
Practical	:	22.5 marks
Internal Assessment	:	2.5 Marks

1. Study of organisation in a flowering plant. Role of shoot apical and root apical meristem (using whole mounts of shot tips of *Hydrilla,* vertical sections of *Coleus* or *Bryophyllum*, and whole mounts of root tips of *Pistia,* or vertical sections of root tips of onion, maize, aerial roots of *Ficus benghalensis* or *Tinospora cordifolia).*

2. Study in the field of a large variety of herbaceous and perennials, both monocotyledonous and dicotyledonous; aquatic, mesophytic, xeromorphic plants, succulents, climbers, twiners, parasites, and trees with varied canopy architecture. Field diary to be prepared.

3. Morphology and anatomy of stems, roots, and leaves using stained hand sections of permanent slides (tapioca, carrot, radish, onion, garlic, turnip, potato, etc.).

4. Study of vascular cambium in stem and root. Secondary growth. Examples of cambial variants and resultant secondary structures.

5. Preparation of world maps to show Vavilov's centres of origin or cultivated plants and Zhukovsky's concept of mega centres.

6. Preparation of maps showing the centres of primary diversity of important economic plants that originated in India and the regions of cultivation of plants introduced from the New World. Africa, Europe, Central Asia, Australia, Malaysia, China.

7. Study of vegetative and reproductive morphology and anatomy of wheat, rice, maize, sorghum, bajra, potato and sugarcane. Also study of starch grains in these plants (except in sugarcane). Epidermal structure of sugarcane stem and leaves. Tests for sucrose.

8. Study of vegetative and floral morphology, and pod development in chickpea and red gram. Familiarization with four fodder legumes (such as berseem, Lucerne, methi, *Clitoria*, guar, cowpea, etc.).

9. Study of structure of oil storing tissues in sectioned seeds of mustard, groundnut and soyabean, and coconut endosperm, using micro-chemical tests.

10. Study of vegetative, floral and fruit morphology of cotton. Ontogeny of cotton fibre in young ovules and tracing the development of cotton fibre. Mature microscopic structure of cotton fibre. Preparation of absorbent cotton by alkali treatment. Staining of cotton with and without mordanting. Study of stem of jute showing origin and development of jute fibre. T.S. coir to show that coir is a fibrovascular bundle.

11. Study of wood anatomy of ring-porous, diffuse porous woods using hand sections or prepared slides. Understanding of wood anatomy using T.S., T.L.S. and R.L.S. in five timber yielding species (*teak*, *Dalbergia, Albizzia, Gmelina, Eucalyptus Cedrus* or *Pinus, Acacia, Shorea, Terminalia, Mangifera, Azadirachta,* or any other depending on the location of the institution). Preparation of a list of 10 fire wood species in your region and their properties.

12. Study of 10 whole plants (live or from herbarium specimens) used as resources of drugs : (*Poppy*, *Rauvolfia, Adhatoda, Emblica officinalis, Terminalia bellirica, T. chebula, Glycyrrhiza, Phyllanthus amarus (P. fraternus*), garlic, *Andrographis paniculata, Catharanthus rosesus, Atropa belladonna, Digitalis purpurea, Aloe barbadense,* or any other important plants of your area.

13. Study of laticiferous system in the stem of *Hevea brasiliensis,* familiarization with whole plants (fresh or herbarium specimens of *Cymbopogon* sp., *Eucalyptus, Jasminum, Geranium, Pandanus* sp., *Rosa, Vetiveria zizaniodes, Santalum album* (or any other local sources of essential oils).

14. Study of the neem plant, *Artemisia annua, Chrysanthemum cinerarifolium (*pyrethrum) from live specimens.

15. Sources of dyes : *Curcuma longa* (turmeric), *Bixa orellana (*annato), *Crocus Sativus, Butea monosperma (*palash), *Indigofera (*indigo), *Lawsonia inermis* (mehndi). Extraction of pigments.

16. Preparation of a list of trees and shrubs used as ornamentals alongwith their popular and scientific names. Seasons of flowering and brief description. Calendar of seasonals grown as bed plants, potted plants, house plants, flowers used for worship or ornamentation. This may be illustrated and presented as a term paper at the time of examination.

ZOOLOGY

B. Sc. (HONOURS) SECOND YEAR EXAMINATION, 2011

PAPER-I : ENVIRONMENTAL BIOLOGY

Max. Marks	:	75
Theory Exam.	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 hrs.

Note : Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

]

UNIT-I

1. **Environment :** Introduction and components.

2. **Ecological Concepts :** Ecosystem, food chain, food web, energy flow.

UNIT-II

3. **Pollution: Types** - Air, water, land, thermal and noise pollution.

(a) Sources of Pollution : Sewage, industrial wastes, silt, fertilizers, detergents, oils, pesticides, smoke, exhaust gases from automobiles, smog, acid rains, radioactive elements etc.

(b) *Hazards of Pollution :* Effects of pollution on plants and animals.

(c) *Pollution in India:* Major polluted rivers and their sources of pollution, air pollution in major cities.

(d) *Control of Pollution:* Primary, secondary and tertiary treatment, standards for the quality of drinking water.

UNIT-III

4. **Wild Life :** Wild life conservation, sanctuaries and national parks, endangered species

UNIT-IV

5. Ecology and Human Welfare :

(a) *Conservation of Natural Resources:* Water, land and minerals.

(b) *Aquaculture:* Pisciculture and management of ponds.

(c) *Forest Ecology :* Forest statistics of India, conservation of forests, management of forests for timber productions.

(d) *Agriculture :*Weed control, crop rotation, green revolution, checking of soil erosion.

(e) *Desert Eradicdtion:* Measures to control the advance of desert.

Suggested Readings :

1. Odum, E.P. : *Fundamentals of Ecology,* W.B. Saunders Co., Philadelphia, 1971.
2. Melanby, K. : *The Biology of Pollution : Studies in Biology,* No. 38, Edward Arnold (Publishers) Ltd., 41, Bedford Square, London,1980.
3. Louis, K. : *River Pollution,* Butterworths, 1969.
4. Dix, H.M. : *Environmental Pollution, Atmosphere, Land, Water and Noise,* (Institution of Environmental Science) Wiley, 1981.
5. Mani, M.S. : *Ecology and Biogeography of India,* Dr. W. Junk B.V. Publishers, The Hague, 1974 .
6. Kendeigh, S.C. : *Ecology with Special Reference to Animals and Man,* New Jersey, Prentice Hall Inc, 1974.

Paper-II : MOLECULAR BIOLOGY

Max. Marks	:	75
Theory Exam.	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 hrs.

Note : Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

1. **Introduction :** Scope and historical account from molecules to cells, prokaryotes to eukaryotes, unicellular to multicellular.

2. **Biomolecules :** Structure and function of carbohydrates, lipids, proteins and nucleic acids.

3. **Carbohydrates :** Classification, structure of cellulose and glycogen.

UNIT-II

4. **Lipids :** Classification, structure of ganglioside and steroids.

5. **Proteins :** Classification, structure of amino acids, primary, secondary and tertiary structure of proteins, structure of hemoglobin and lysozyme.

6. **Nucleic Acids :** DNA. What genes are made of ? Detailed structure of DNA Molecule, replication, RNA - various types, transcription, translation and involvement ofRNAs in protein synthesis.

UNIT-III

7. **Mutations :** Types of mutations, physical and chemical mutagens and their mode of action on DNA.

8. **Bioenergetics :** Types of energy and their transformation, first and second law of thermodynamics, factors governing chemical reactions, generation and function of ATP, biocatalysts; role of enzymes in cellular reactions. Types of major enzymes.

9. **Regulation of Gene Expression :** Concept of operon, induction and repression in Prokaryotes, Lac operon.

UNIT-IV

10. **Molecules of Cellular Defence :** Structure and function of antibody, major histocompatibility antigens, HLA system. Genetic basis of antibody diversity.

11. **Molecular Basis of Inborn Errors of Metabolism :** Genetic basis of carbohydrates and amino acid metabolism, galactesemia and Phenylketonuria.

12. **Recombinant DNA :** Restriction endonucleases, Plasmids, cloning of DNA, use of recombinan DNA technology.

Books Recommended :

1. Alberts, B. Bray, D., Lewis, J., Raff, M., Robert, K. and Watson, J.D. : *Molecular Biology of the Cell,* Garland Publications, New York,1983.
2. DeRobertis, E.M.F. Jr. : *Essentials of Cell and Molecular Biology,* Halt-Saunders, Japan, 1981.
3. DeRobertis, E.D.P. and DeRobertis, E.M.F. : *Cell and Molecular Biology,* 8th Edition, Halt Saunders, Tokyo, 1987.
4. Gardner, E.J., Simmons, M.J. and Snustad, D.P. : *Principles of Genetics,* John Wiley and Sons, New York, 2004.
5. Power , C.B. : *Cell Biology,* Himalayan, Publishing House, Bombay,1981.
6. Schleif, R. : *Genetics and Molecular Biology,* Addison-Wesley, Reading U.K., 1986.
7. Villee, C.A., Selomon, E.P. and Davis, P.W. : *Biology,* Saunders College Publishing, New York, 1985.
8. Watson, J.D. : *Molecular Biology of the Gene,* 4th Edition, Vols. I & II, Narosa Book, New Delhi, 1987.
9. Barry, J.M. : *Molecular Biology, Genes and Chemical Control of Living Cells,* Prentice Hall Inc., New Jersey, 1964.

PRACTICALS

Practicals based on both theory papers.

I. Determination of Physico-chemical factors in an aquatic habitat :

1. Temperature
2. Light
3. pH
4. Dissolved Oxygen
5. Carbon Dioxide
6. BOD
7. Alkalinity
8. NO_3
9. PO_4

II. Analysis of polluted water : Sewage/industrial effluent for physico-chemical factors as stated in No. I.

III. Biota of pond water and sewage/industrial effluent.

IV. Effect of light and gravity on organism.

V. Study of littoral, limnetic, profundal communities.

VI. Analysis of the variations in protein contents within the organ of an animal and in specimens of same species from different areas.

VII. Analysis of the species variations in amino acids requirements.

VIII. Demonstration of carbohydrates and proteins in a cell (Cytochemically).

IX. Study of nucleic acids in tissues (Cytochemically).

X. Idea about viruses and bacteria through charts etc.

XI. Preparation and study of plant/grasshopper chromosomes.

The scheme for the conduct of Practical Examination.

Based on Theory Papers I & II (Honours)

Max. Marks	:	50
Practical Exam.	:	45 marks
Internal Assessment	:	5 marks
Time	:	4 hrs.

I. Analyse the variations in protein contents in the organs of a given specimen. Write the procedure and show the reading to the examiner. (6)

II. Determine following physico-chemical factors in given water sample from a lake. Write the procedures followed. (6)

III. Demonstrate cytochemically carbohydrates/protein in an eukaryotic cell. Write the procedure and show the preparation to the examiner. (5)

IV. Record biota in the given sewage sample. Draw labelled sketches & demonstrate these to the examiner. (4)

V. Prepare permanent stained mount of the chromosomes of material provided. (4)

VI.	Examine water sample for biotic communities from particular zone of lake, sketch and demonstrate.	(4)
VII.	Identify the given chart/diagram. Write a brief note on it.	(2)
VIII.	Study the effect of _____ on the given animal. Write the procedure followed.	(4)
IX.	Viva-Voce.	(5)
X.	Note book and Charts.	(5)

Published by : **Professor S.S. Bari, Registrar, Panjab University, Chandigarh.**